

RENAULT NISSAN DESIGN SPECIFICATION
(RNDS)

Vehicle generic specifications

Design specification

RNDS-C-00117 v3.0

RENAULT STANDARD

(GDNormes)

SPECIFICATION

SHEET

31-08-874 --H

NISSAN DESIGN SPECIFICATION

(NDS)

SPECIFICATION

Issued: 2017-03-22



(030; 038; 048; 054; 064; 070; 089; 091, 094)



Title: **TIRES**

Confidentiality level: Renault Nissan Confidential C

Signature (Alliance Single Technical Leader)

Shinji Matsumoto

Signature (Renault validator)

Fabrice GARCIA

Signature (Nissan validator)

kugo shiro

Update list :

- PPR13 annex : HJD = 14,1 mm ieo 14 mm in the example
- CRS to be removed
- Recycling: 00-10-060 => RNES is existing?
- 9 S sigles removed
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Revisions

□□	□□	xxx	20XX.XX.XX	□□□□□□			
				□□□□□□			
H.Harada	C.Rigault T.Mizoguchi	DEA-SC XJS	2019.03.28	Safety converged	3.0	3.0	2.0
S.Matsumoto	V.Masi T.Mizoguchi	DEA-SC XJS	2017.03.22	Tests specifications	2.0	2.0	1.0
				Sigles evolutions			
				Include spare tire requirement			
S.Matsumoto	V.Masi T.Mizoguchi	DEA-SC XJS	2016.03.29	Newly established	1.0	1.0	N
APPROVED BY	SIGN	SECT	APPROVED DATE	ALTERATION	CHG RNDS	Ver GDNormes	CHG NDS

RENAULT-NISSAN

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Foreword

This document intend to replace current Renaul and Nissan standards and establish a new common specification for tires.

Renault	Issued by	: Vincent Masi
	Validated by	: Fabrice Garcia
NISSAN	Issued by	: Taishi Mizoguchi
	Validated by	: Shiro KUGO

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PRODUCT SPECIFICATION

1. Scope

The purpose of this specification is to define the characteristics required for the tyres intended for the wheels of passenger and light commercial vehicles.

2. Normative references

These normative references apply the latest version.

Standards	: 00-10-415; 00 10 050 ; 03-50-000 ; 41-02-001.
Test Procedures	: 31-08-031, 31-08-871, 31 08 878, 31 08 882, 37 00 041, CPGEX.
Test methods	: D47 1053, D41 1099, D47 1100; D47 1924.
ISO standards	: 10191 ; 18164 ; 28580
WDK standards	: 107 ; 109 ; 110 ; 113 ; 116 ; 221.
European regulations	: ECE 30 , 54, 117.
EEC directives	: 92/23 ; 2005/69/CE.

3. Terms and definition

For the purpose of this standard, the following terms and definitions apply.

- 3.1 FP - Main Function (or) Principle Function which are related to the direct function of the part
- 3.2 FC - Constraint Function which are related to the environment where the part needs to be compliant
- 3.3 TPG – Tochigi Proving Ground which is related to the dynamic vehicle testing location
- 3.4 CPVS – Containment Program for Vehicle Shipment which is related to the standard followed for evaluation

4. Symbols and abbreviated terms

- 4.1 HCPP – Product / Process Characteristic Hierarchy
- 4.2 ETRTO - European Tyre and Rim Technical Organisation
- 4.3 ECE – Economic Commission of Europe
- 4.4 JIS – Japan Industrial Standard
- 4.5 JASO – Japanese Automobile Standards Organisation
- 4.6 PFE – Elementary Function Leader (refer to Engineer in charge of Tire development)
- 4.7 IMDS – International Material Data System
- 4.8 ABS – Anti-lock Braking System

5. TIRES FUNCTIONAL DEFINITION

5.1. PRODUCT LIFE CYCLE

LIFE PHASES	studied	Not studied
LOGISTICS – Packaging – Storage 1 – Transportation – Acceptance – Storage 2 – Unpackaging – Delivery rate		X X X X X X X
ASSEMBLY – Preparation before assembly – Assembly on vehicle – Startup in plant and inspections	X X X	
DELIVERY OF THE VEHICLE – Vehicle storage – Vehicle transportation – Vehicle placed at disposal		X X X
UTILISATION BY FINAL CUSTOMER – stopped – running – braking – subject to impacts – parking – cleaning (brushes, high pressure jet, manual cleaning)	X X X X X X	

5.1.1. CHARACTERIZATION OF FUNCTIONS (verb + element describing the external environment-EME)

Explanations:

Criterion : figures or qualities to define the performance expected from the product.

Level : rank on criteria measurement scale.

Test Method: method allowing to check that the level has been reached.

Flexibility (indicates whether a criterion can be negotiated or whether it is mandatory) :

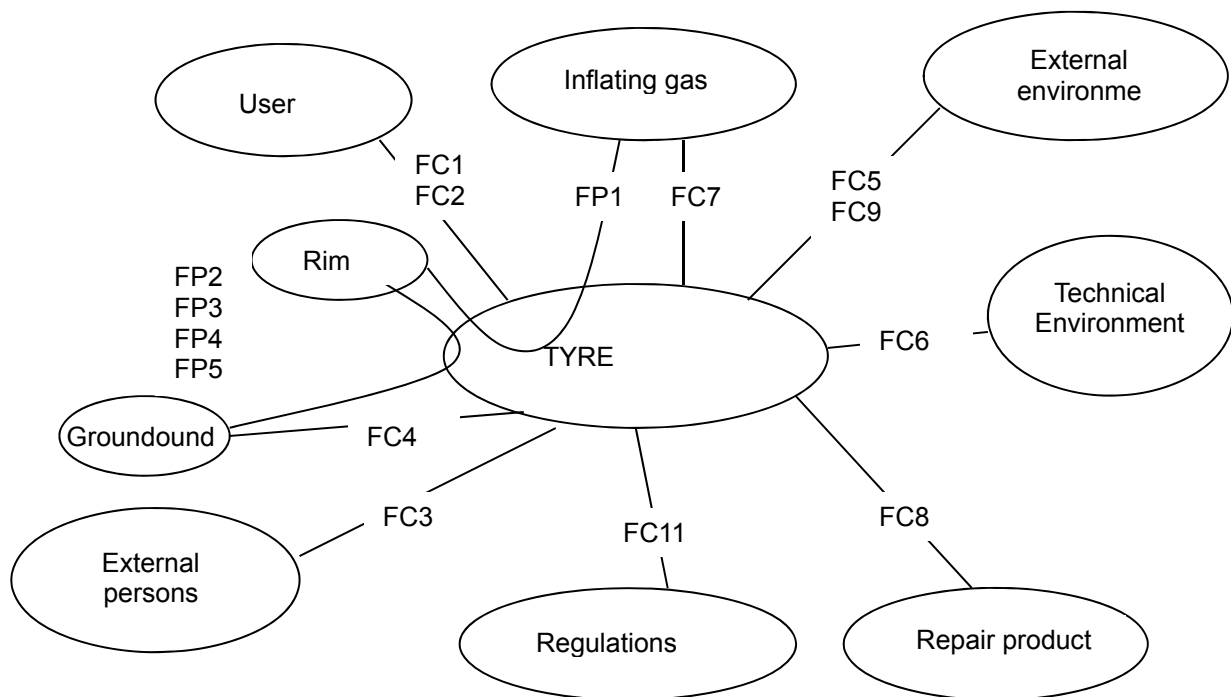
- F0 : no flexibility, mandatory ,
- F1 : low flexibility, less negotiable.
- F2 : good flexibility, negotiable,
- F3 : very good flexibility, very negotiable.

HCPP : Product /Process Characteristic Hierarchy. The Rate of Non Compliance (TNC) associated with each severity class are indicated in the Table of Disturbed Service Functions (FSP) figuring in [Annex D](#)

Person in charge : person in charge of checking that the level has been respected.

5.2 LIFE CYCLES:

5.2.1 Life cycle phase: "Using by the Customer"



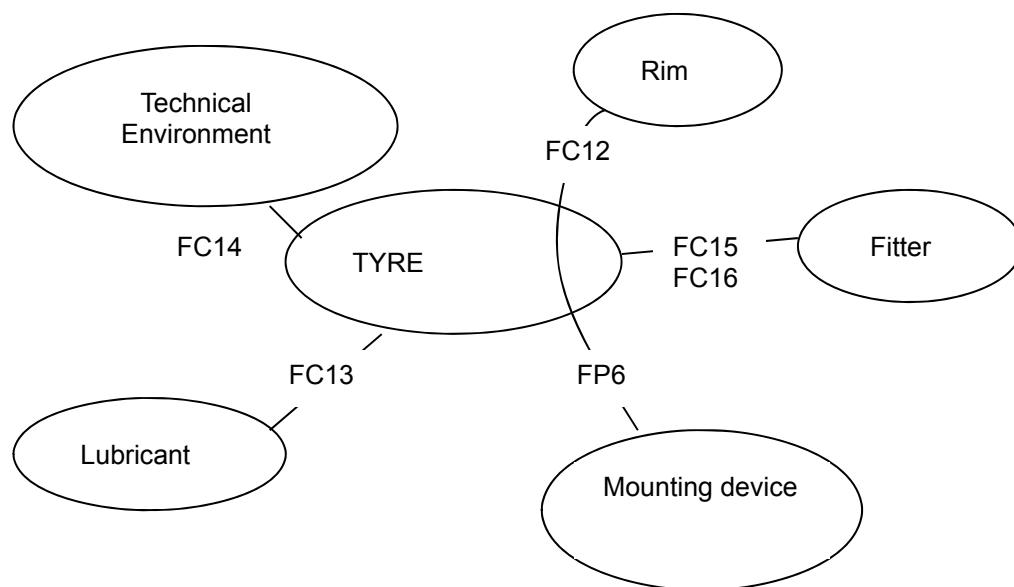
Functions to be ensured

FP : Main Function and **FC**: Constraint Function

- FP1: The tyre must bear the vehicle load transmitted by the rim, supported by the inflating gas.
- FP2: The tyre shall transmit the forces between the rim & ground (acceleration, braking, cornering, etc.).
- FP3: The tyre must transmit static electricity from the rim to the ground.
- FP4: The tyre must filter the forces transmitted by the ground to the rim (passenger comfort, and preserve the integrity of the parts stressed by it).
- FP5: The tyre participates in the energy performance of the vehicle.
- FC1: The tyre must comply with the user (running noise heard by the occupants of the vehicle from inside the vehicle, ...).
- FC2: The tyre must give information to the user (marking, wear indicator, certification, etc.).
- FC3: The tyre must respect the people outside the vehicle (external noise, appearance, etc...).
- FC4: The tyre must resist the ground (perforation, heat, soil type, etc..) .
- FC5: The tyre must resist the external environment (washing product, hydrocarbons, etc.).
- FC6: The tyre must respect its technical environment (body, axle, wheel cover, speedometer, ABS system...).
- FC7: The tyre must resist the inflating gas.

- FC8: The tyre must resist the repairing product.
- FC9: The tyre must respect the external environment (non-polluting material, ...).
- FC11: The tyre must comply with the regulations in force.

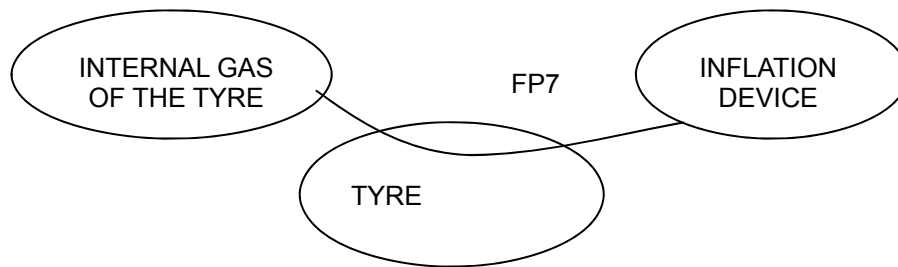
5.2.2 Life cycle phase: "ASSEMBLY"



Functions to be ensured

- FP6: The tyre must be installed on the wheel with the mounting device
- FC12: The tyre must be installed on the rim.
- FC13: The tyre shall be compatible with the lubricant used.
- FC14: The tyre must comply with its technical environment (balancing weight, roller bench, etc.).
- FC15: The tyre must inform the fitter about its direction/orientation of installation.
- FC16: The tyre must comply with the fitter.

5.2.3 Life cycle phase: "Tyre inflation"



Functions to be ensured

- FP7 : The tyre shall allow pressurization and the adjustment of pressure by the inflation system.

5.3 CHARACTERISATION OF FUNCTIONS

5.3.1 Life cycle phase: "Use by the Customer"

5.3.1.0 UCEps : Unwanted Customer Events potentially safety related

The supplier must take into account the list of Events potentially safety below, relevant for the tire. These FCEps are extracted from the common Alliance list.

- FECps n°11: Loss of control: due to an absence of braking or an unexpected loss of the braking or a loss of assistance of braking not indicated, including due to a driving with parking brake tightened not indicated. (during the driving)
- FECps n°18: Loss of control: due to a sudden loss of the pressure or the explosion of a tire (except aggression by elements outside the vehicle). (during the driving)

The supplier is in charge to:

- complete the safety events list based on its experience, expertise or analyses.,
- give to Renault or Nissan the details linked to these events and the use of his product on vehicle,
- to be able to attest how the safety events have been taken into account in product design and manufacturing process of the tire. (Design Failure Mode and Effects Analysis – see ANPQP requirements)
- provide demonstration elements (refer §7 RELIABILITY – DURABILITY)

5.3.1.1 FP1 : The tyre must bear the vehicle load transmitted by the rim, supported by the inflating gas.

Characterization of the elements of the external environment of use:

User: vehicle driver

Rim : Rim made of steel sheet or stamped aluminium which is coated of protection (Electrostatic painting or paint)

Light alloy rim moulded or forged which is coated with protection (paint + varnish).

The definition of the tyre and wheel interface is compliant with ETRTO.

The dimensions of the rims to be used for testing are indicated in the functional file of the tyre

The rim used for the tests will always be the one intended for the vehicle. Its definition can change over time depending on the progress of the vehicle in the project. The rim will be provided by Renault-Nissan, if not available, a rim of similar size in terms of width and wheel offset must be used in accordance with Renault-Nissan. If some tests require the use of different size of rim, it must be stated in the test report with the results.

The minimum and maximum inflating pressure at ambient temperature If the supplier has recommendations different from those described in ETRTO, he must inform about them.

Reference pressure (inflation pressure for testing): Unless otherwise indicated, use the ETRTO reference pressure indicated by "INFLATION PRESSURE" in the tables of the series "METRIC DESIGNATION" of tyres.

Maximum speed of the vehicle: The tyre delivered by the manufacturer must have a speed index compatible with the maximum speed of the vehicle on which it will be mounted.

Maximum loads of the vehicle that the front and rear axles can bear: The tyre delivered by the manufacturer must have a load index compatible with the loads of the vehicle on which it will be mounted.

Internal gas : pressurised ambient air, nitrogen, helium (possible in after-sales service).

Technical Environment

Wheel arch, elements of axles (suspension etc.), wheel cover, balancing weight, compartment of spare wheel (inside the vehicle in the trunk, outside under the body) in a wire carrier or maintained by a winch, or fixed to the body outside the vehicle).

Additional attachment device:

The additional attachment device must be in compliance with the French and European standards. The size of the chain planned on the vehicle is mentioned in the functional file of the tyre.

Repairing product: Inflating kit (reference indicated in the functional file of the tyre).

Environment:

- Temperatures: between - 40 °C and 60 °C (except for prescriptions mentioned in the functional file of the tyre).
- Fluids: hydrocarbons, snow clearing solution, brake fluid, washing fluids, liquid from the tyre repair kit.
- Ozone.
- Gravel, dust, water, mud, snow, etc.

External environment: Ground more or less regular (asphalt road, concrete road, paved road, dirt tracks, etc.), pavements, etc.

Characterization of the assessment criteria:

Details on test conditions:

Subject to the compliance by the manufacturer with the tyre dimension (including load and speed indices), the assessment of the tyre in the function FP1 is covered by the other main functions and constraints described in this specification.

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Sealing at room temperature	Loss of pressure less than 0.1 bar after 1 month of storage	6 tyres assembled on wheel rim Initial pressure = ETRTO reference pressure. Checking of the pressure without disconnecting the pressure gauge from the valve T = 23 °C ± 5°C.	F0		Supplier
Load/Speed performance test	The tyre must have no degradation (separation or pulling of the tread, cracking of folds, etc.).	Minimum 1 mini tyre mounted on the rim of vehicle (1) E / ECE / 30 for Passenger car E / ECE / 54 for Commercial Vehicles (ECE R30/R54 marking is required for all destination)	F0		Supplier
Tubes specifications of tyres with tube	Tubes specifications shall comply with JIS D4207	Supplier Drawing	F2		Supplier

(1): Test necessary for the certification of tyre, to be conducted on the final fabrication selected during various tests & declared as such by Renault / Nissan

5.3.1.2 FP2 : The tyre must transmit the forces between the wheel rim and the ground

FP4: THE tyre **must filter the forces transmitted** by the ground to the wheel rim

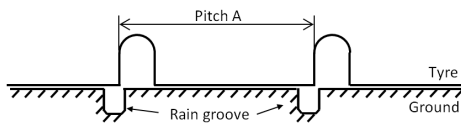
Characterization of elements of the external environment of use: see FP1.

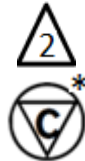
The supplier must be able send an F-Tire model of the tyre defined in the functional file in case of request expressed by Renault. This is used to create simulations of driving on endurance track. See [Annex A](#)

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Comfort, Ride and Handling	The performance level must be upper or equal to the objective defined by the Customer Prestation Pilot during the job sharing rating grid	4 tyres mounted on the vehicle. Subjective evaluation on vehicle compared to a reference defined tire. Applicable test procedures : 3108882 & Nissan Method	F1		RENAULT / NISSAN
Sensitivity to lateral tilt (Acceleration at steady speed).	Steering Wheel Mass correction (g); steering wheel torque (N.m) and the judgement of acceleration (slow or fast) must be acceptable.	4 tyres mounted on the vehicle. Subjective evaluation on vehicle compared to a reference defined tire Applicable test procedures : 3700041 & Nissan Method	F1		RENAULT / NISSAN
Severe lane-change manoeuvre (Bayonet)	The performance level must be upper or equal to the objective defined by the Customer Prestation Pilot during the job sharing rating grid	4 tyres mounted on the vehicle. Subjective evaluation on vehicle compared to a reference defined tire Applicable test procedures : ISO3888-1 and 2 & Nissan Method	F0		RENAULT / NISSAN
Pulling under torque	The performance level must be upper or equal to the objective defined by the Customer Prestation Pilot during the job sharing rating	Subjective evaluation on vehicle compared to a reference defined tire	F1		RENAULT / NISSAN

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Static & dynamic characteristics (1) Cornering Stiffness Camber stiffness Sway stiffness Static Radial stiffness Tyre castor Tyre/ground friction torque Vertical spring constant N/mm Lateral spring constant, N/mm MAX cornering force kN Cornering power kN/° Self-alignment torque power N-m/° Camber thrust coefficient N/°	If applicable specific targets are defined for each project. All the measurements of static and dynamic characteristics must be mentioned on the drawing (On the title block selected by the project) Mass production allowed tolerance for these characteristics : +/-3%	Applicable test procedures : See Annex B 31 08 878 & Nissan Method & Standard drawings	F1		Supplier
Rim Slippage	Tire rim slippage after testing under the following requirements shall be less than 30 mm in the circumferential direction of the rim flange.	See Annex H	F1		Supplier
Bead unseating test	Value of force according to the nominal width of the tyre (W), more than:	Minimum 2 tyres on the test bench. ISO 10191	F0		Supplier
	For $W < 160 \text{ mm}$ => $F > 6\,670 \text{ N}$ For $160 \leq W \leq 205 \text{ mm}$ => $F > 8\,890 \text{ N}$ For $W \geq 205 \text{ mm}$ => $F > 11\,120 \text{ N}$	Test on the wheel rim of the project			

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Dry braking Longitudinal adhesion on dry ground	Braking Distance should be be better or equal to the weighted target relative to the reference tire. The evolution of stopping distance measurements (% ofstabilization) on the first 15 applications of the brake(on dry ground) from 130 km/h to 0 km/h must be less than 4 %.	4 tyres mounted on the vehicle with ABS. 100 km/h to 0 km/h Applicable test procedures : 31-08-031	F0		Supplier Renault / Nissan
Wet braking Longitudinal adhesion on wet ground	Braking Distance should be better or equal to the weighted target relative to the reference Tire	4 tyres mounted on the vehicle with ABS. 80 km/h to 0 km/h Applicable test procedures : 31-08-031	F0		Supplier Renault / Nissan
Wet Grip	Tyre should comply with wet grip requirements as included in ECE117 Conformity to be indicated on drawing	According ECE117			Supplier
Quality Requirement for North America	Actual performance of the tire shall be grade A. However, grade B is acceptable when agreed upon with the Renault/Nissan design section in charge as an exception	UTQGS TRACTION performance	F1		Supplier
	The pattern pitch A in the tire width direction shall not fall within ± 1.5 mm of an integer multiple of 19 mm. 	None	F1		Supplier
Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge

Uniformity Radial force variation (RFV) - Overall (Peak to Peak) 1st Harmonic 2nd Harmonic Lateral force Variation (LFV) - Overall	Technical requirements defined for each project. See Spec tender or compliance matrix for targets.	Applicable test conditions to be precised project by project. ISO 13326 (OR) WDK109 With load case WDK109 + with vehicle conditions Inflation pressure: 200 kpa Wheel load & Rim width: ref WDK109 sheet 3	F1		Supplier
Conicity	Technical requirements defined for each project. See Spec tender or compliance matrix for targets.	Applicable test conditions to be precised project by project. ISO 13326 (OR) WDK109 With load case WDK109 + with vehicle conditions Inflation pressure: 200 kpa Wheel load & Rim width: ref WDK109 sheet 3	F2		Supplier
Flat Spot	Tyre must comply with flat spot requirements : a) Short term requirements b) Long term requirements	See Annex G	F2		Supplier
Buffing to improve uniformity	a) Amount of buffing - 0.5 mm or less b) Reduction of RFV (Peak to Peak) – 40N or less c) The appearance shall not be deteriorated after being buffed	None	F0		Supplier
Static unbalance	Technical requirements defined for each project. See Spec tender or compliance matrix for targets.	Applicable test conditions to be precised project by project ISO 13326 (OR) WDK113	F1		Supplier

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Radial runout	Technical requirements defined for each project. See Spec tender or compliance matrix for targets.	ISO 13326 (OR) WDK109	F3		Supplier
Residual cornering force (RCF)	Technical requirements defined for each project. See Spec tender or compliance matrix for targets.	Refer to Annex B	F1		Supplier
Residual torque of Ply steer (Cra)	Technical requirements defined for each project. See Spec tender or compliance matrix for targets.	According to the load case WDK 109. Refer to Annex B Minimum 4 tyres	F1		Supplier

(1): The static and dynamic characteristics will be measured on the final fabrication selected during the various tests and declared as such by Renault/Nissan.. All the measurements will be mentioned on the tyre drawing to be sent by the supplier.

*: Specific to 1st and 2nd Harmonic measurement

5.3.1.3 FP3 : The tyre must transmit static electricity from the rim to the ground.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Electrical Resistivity Electric resistance between Tire Bead and Tread Center	Resistivity of the tyre must be less than 10^8 Ohms	WDK 110 Minimum 3 tyres mounted on the steel wheel rim of the vehicle (If they exist).	F0		Supplier
Electric resistance between Tire Beads (along the circumference)	Resistivity of the tyre must be more than 10^4 Ohms	See Annex C	F0		Supplier

5.3.1.4 FP5: The tyre participates in the energy performance of the vehicle.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Rolling resistance	Level of rolling resistance in compliance with the objective values of the functional file of the tyre	ISO 28580 value recalibrated according to the European virtual laboratory 2 tyres assembled on nominal rim ETRTO or rim of the project according to request.	F0	 (070; 038; 030, 094)	Supplier
	All the measurements of rolling resistance must be mentioned on the drawing (On the title block selected by the project)	See Annex B	F3		Supplier
	Tolerance on Rolling Resistance to be applied in Mass production : +/-0,3 Kg/t	ISO 28580	F3		Supplier

5.3.1.5 FC1: The tyre must comply with the user.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Visual appearance of the tyre in delivery conditions	The supplier will have to present, tyres sorted and representative of what he considers as manufacturing acceptance limits for the dimension and the brand line which he proposes V1 - Strong criticism Indentation upper to 1.4mm V2 - Potentially open to criticism 0.9 mm < Indentation <=1.4 mm V3 - Non critical : Indentation <0.9 mm The concave on the side wall of the tire shall be 0.8 mm @400 kPa or less as a guideline. The tire should not show any visible texture due to curing/reworking.	A check of the general appearance will be made and validated by the representatives of the perceived Quality, in particular on the possible indentations of the sidewall of tyres.	F3		Supplier Renault / Nissan
Acoustics a) Internal running noise on rough road b) Internal running noise on smooth road	Noise levels corresponding to the objective values of the functional file of the tyre. Noise value comparizon versus reference tyre for the following frequency ranges. [15 to 60 Hz] [60 to 250 Hz] [250 to 500 Hz] [500 to 8000 Hz] Global [15 to 8000 Hz]	4 tyres mounted on the vehicle. Tests according methods described in project documentation (RFQ Package).	F1		Supplier

The supplier will have to send a Finite Element Model of the tyre selected during the various tests and declared as such by Renault. The FEA model will have to comply with specifications provided by Renault/Nissan.

5.3.1.6 FC2: The tyre must give information to the user (marking, wear indicator, certification, etc.).

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Marking Dimension and characteristics of the tyre (speed and load indexes, direction of rotation, type of structure, use, etc.	The designation must be in compliance with the markings required by the regulations. Definitions and positions to be indicated on the drawing. ECE30/54 regulation requirements (markings and certificates) must be applied for all Tyres independently of commercialization countries.	Regulations : See the project functional file to identify the regulations defined according to the geographical zones : Example: 92/23/CE Or E/ECE/117 E / ECE / 30 for passenger cars E / ECE / 54 for commercial vehicles	F0	 (070, 094)	Supplier
Wear indicator	Minimum 6 indicators per groove with an identification mark on the sidewall of the tyre. Definitions and positions to be also indicated on the drawing. Height of indicator: 1.6 mm +0.6 mm/- 0mm;(for summer tires)	control report on 1 tyre	F0	 (070)	Supplier

5.3.1.7 FC3: The tyre must comply with the people outside the vehicle.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
External running noise Average value at 50 km/h (coeff. B) Value without rounding-off or deduction of the tolerance dB	Noise level in compliance with the objective value of the functional file of the tyre	Coast down measure according - ECE 117.02 On circuit - ISO 10844 Pressure according to ECE117.02, para. 2.5.3 Min. 2 passages at 40, 50, 60, 70, 80 kph Calculation of the coefficients A and B of logarithmic interpolation according to the law: $L_p = A \cdot \log_{10}(V_{kph}/50) + B$ Application of the correction in temperature, according to ECE 117.02, para. 4.3 Values without rounding-off or tolerance dB (ECE 117.02 paragraphs 4.4 and 4.5) See Project documentation for details concerning tests specifications.	F1	 (070; 054)	Supplier
Regulation requirement	The noise level must be less than the level required by the legislation in force.	Regulations ECE/117	F0	 (070)	Supplier

5.3.1.8 FC4: The tyre must resist the ground.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Wear 4 tyres mounted on the vehicle	Wear level should be better or equal to the weighted target relative to the reference Tire/vehicle.	See : RNDS-C-00093	F1		Supplier
Tire strength and robustness 4 tyres mounted on the vehicle.	No loss of elements of the tread, bulges, cuts, deformation, etc. with respect to constraints of driving conditions made more severe: pothole, manhole covers, speed bumps, trenches, water ford, gravel, etc.	Inspection after wear test and internal vehicle durability testing (example CPVS vehicle tests)	F1		Supplier
Quality Requirement for North America - Wear	The sipes positioned in the central area of the tread should be still visible at 50% of wear life (i.e. 50% of net groove depth – including the Tread wear indicator	None	F1		Supplier
Tyre Durability	Tyre must fulfill bench test requirement at least equivalents to ECE30/54 or FMVSS 139 requirements. Assesment of tyre durability compatibility with market technical requirements	Used Test procedure equivalence to be demonstrated by supplier & indicated on drawing Supplier must provide validation or tyre performance in accordance with all marketing countries of the project	F0		Supplier

5.3.1.9 FC5: The tyre has to resist the external environment.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Initial properties in the tension Minimum 4 samples per type of material composing the side wall and the tread of a tyre.	Tensile test in an initial state Resistance to rupture R _r (in MPa) Elongation at rupture A _r (in %) No requirement, R _r and A _r are the references to determine the variations after ageing.	D41 1099	F1		Supplier
Resistance to hydrocarbons (1) Minimum 4 samples per type of material composing the side wall and the tread of a tyre	Immersed in the diesel fuel: 15s 23°C, then storage 24 @ 23°C. Immersed in petrol: 15s 23°C, then storage 24 @ 23°C. No rupture, crack or swelling must be observed Tensile test on old samples (immersion + storage) Variation R _r / initial = 20 % Variation A _r / initial = 30 %	D471924 Fluids O and Y according to standard 03-50-000 D41 1099	F1		Supplier
Resistance to brake fluid (1) Minimum 4 samples per type of material composing the side wall and the tread of a tyre.	Immersed in the fluid for 15s 23°C, then stored 24 hours @23°C. No rupture, crack or swelling must be observed after the test	D471924 Fluid type 3 according to standard 41-02-001 D41 1099	F1		Supplier

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Resistance to zinc chloride (1) Minimum 4 samples per type of material composing the side wall and the tread of a tyre.	Immersed in the fluid (Fluid constituted in weight by 50 % of chloride and the rest water) for 24 hours 23°C, then stored 24 hours @23°C. No rupture, crack or swelling must be observed after the test Tensile test on old samples (immersion + storage) Variation R_r / initial = 20 % Variation A_r / initial = 30 %	D471924 D41 1099	F1		Supplier
Resistance to calcium chloride (1) Minimum 4 samples per type of material composing the side wall and the tread of a tyre.	Immersed in the fluid (Fluid constituted in weight by 50 % of chloride and the rest water) for 24 hours 23°C, then stored 24 hours @23°C. No rupture, crack or swelling must be observed after the test Tensile test on old samples (immersion + storage) Variation R_r / initial = 20 % Variation A_r / initial = 30 %	D471924 D41 1099	F1		Supplier
Heat resistance (1) Minimum 4 samples per type of material composing the side wall and the tread of a tyre.	After exposure for 7 days @ 70°C and return to 23°C. Tensile test on old samples Variation R_r / initial = 20 % Variation A_r / initial = 30 %	D47 1053 D41 1099	F1		Supplier

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Resistance to ozone (1) Minimum 4 samples per type of material composing the side wall and the tread of a tyre.	Concentration of 200 ppcm during 24 hours 40°C. No cracking or fissures observed after test	D47 1100	F1		Supplier

(1) If a different procedure exists at the supplier's, its equivalence must be demonstrated. After approval from Renault, the supplier can use it. The validation is made in a generic way for the used materials and not for every tyre.

5.3.1.10 FC6: The tyre must comply with its technical environment.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Maximum dimensional profile Tyre mounted on the rim of the vehicle.	External dimensions of the tire (width and diameter) considered with production dispersions have to be include in the maximum profile defined by the PPR13 see Annex E Tyre real profile and comparizon with Min and Max requirements must be included on supplier drawing.	WDK 221 and inflation pressure of Regulation No. 30&54 E/ECE/324 (1)	F0	 (070)	Supplier

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Minimum dimensional profile Tyre mounted on the rim of the vehicle.	External dimensions of the tyre (width and diameter) considered with production dispersions must be, in their minimum tolerance, compliant with the nominal values defined by the ETRTO. Tyre real profile and comparizon with Min and Max requirements must be included on supplier drawing.	WDK 221 and inflation pressure of Regulation No. 30&54 E/ECE/324 (1)	F1		Supplier
Tolerances on Tyres dimensions	MAXIMUM Tolerances to be applied on mass production & indicated on drawing External diameter : Nominal +/- 3mm Tyre Width : Nominal +/- 3mm The capability must be demonstrated according to the requirements of the ability contract Tolerances must be adjusted if needed to comply with Max & Min profiles.	WDK 221 and inflation pressure of Regulation No. 30&54 E/ECE/324 (1)	F1		Supplier
Circumference of tyre. Tyre mounted on the rim of the vehicle.	The nominal value of the rolling circumference of the tyre has to be in accordance with the ETRTO. The tolerance has to be in compliance with the WDK 128 (+1,5% -2, 5%). (2):	WDK 107	F1	 (048, 054)	Supplier
Static loaded Radius Tyre mounted on the rim of the vehicle.	The value of the static loaded radius of the tyre has to be in accordance with the tolerance of WDK 128 ($\pm 2\%$). (3):	WDK 107	F1		Supplier
Roll of protection of wheel rim. (for wheel protection analysis) Tyre and drawing	If a rim guard is intended on the tyre, a complete description must be provided in supplier drawing	See Annex F	F1		Supplier
Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Profile of the tyre under vertical loads (for wheel protection analysis)	Provision of a drawing in the TIFF format	See Annex F	F1	4	Supplier

Min Load of vehicle and specified max pressure Max Load of vehicle and Mini pressure specified for the vehicle					
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- (1) For information: 1.8 bar for tyres with Standard Load ; 2.2 bar for Reinforced tyres or Extra Load and 4.2 bars for tyres with T type temporary use. For the commercial vehicles refer to the ETRTO.
- (2) It is also requested to measure the rolling circumference in the pressures and the load defined to the functional document of the tyre according to the speed (from 20 km/h to maximal speed of the tyre).
- (3) It is also requested to measure the radius loaded to the pressures and the load defined to the functional folder of the tyre according to the load (from 0 kg to the max load of the tyre).

5.3.1.11 FC7: The tyre must resist the inflating gas.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Behaviour to the nitrogen / Helium Sealing at ambient temperature 2 tyres assembled on wheel rim.	Loss of pressure lower than 0,1 bar after 1 month of storage. The supplier has to indicate any incompatibility with these gases.	Initial pressure Benchmark pressure of ETRTO Continuous control of the pressure without disconnecting the manometer of the valve. T = 23 °C ± 5°C	F1		Supplier

5.3.1.12 FC8: The tyre must resist the repairing product.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
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Resistance to repairing fluid : 2 tyres assembled on wheel rim	In the recommended practices of use, the supplier will have to indicate if it is the case, the limitations of use of product of repair on its tyres.	None	F1		Supplier
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5.3.1.13 FC9: The tyre has to respect the outer environment.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Component materials the tyre. completion of the material data sheet IMDS (mandatory for RENAULT and on request for NISSAN).	No substances with prohibited use No aromatic hydrocarbons.	Standard 00 10 050 Guideline 2005 / 69 / CE (Guideline 2000 / 53 / CE) REACH	F0		Supplier
Recycling	The manufacturer will be organized to manage the worn tyres and participate actively with the collection and valuation sectors of the worn tyres.	00-10-060	F0		Supplier
Weight	The mass has to be in accordance with the objective value of the Spec. Tender / Functional file of the tyre +/-3 %.	Capability must be demonstrated according to requirement of the ability contract	F1		Supplier

5.3.1.14 FC11: The tyre must comply with the regulations in force.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
The supplier must comply to perform all that would be necessary and tests under its responsibility in order to obtain the necessary approval for the industrialization of its tyre then for its use in these various countries. The supplier will have to provide certification numbers of his product and supply the certificates at the request of RENAULT/NISSAN Commercialization countries to be considered are included in Spec. Tender / Functional files of the project.	The tyre has to be in accordance with all the set of all criteria mentioned in the regulations and/or the guidelines. Information to be shown on the supplier drawing according to the format of the standard drawing.	According to regulatory requirements	F0	 (070)	Supplier

5.3.2 LIFE PHASE “Assembly”

5.3.2.0 UCEps : Unwanted Customer Events potentially safety related

The supplier must take into account the list of Events potentially safety below, relevant for the tire. These FCEps are extracted from the common Alliance list.

- FECps n°18: Loss of control: due to a sudden loss of the pressure or the explosion of a tire (except aggression by elements outside the vehicle). (during the driving)

The supplier is in charge to:

- complete the safety events list based on its experience, expertise or analyses.,
- give to Renault or Nissan the details linked to these events and the use of his product on vehicle,
- to be able to attest how the safety events have been taken into account in product design and manufacturing process of the tire. (Design Failure Mode and Effects Analysis – see ANPQP requirements)
- provide demonstration elements (refer §7 RELIABILITY – DURABILITY)

5.3.2.1 FP6 : The tyre must be set up on the wheel with the mounting device

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Assembly of the tyre on wheel rims. 4 tests mini + industrial tests.	The assembly must be considered satisfactory by the assembly plant on all the wheel rims of the project..	Fittability test in Plant	F0		Supplier Renault / Nissan

5.3.2.2 FC12: The tyre must be installed on the rim.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Bead base clamping Force. Test of measures of effort on 6 minimum tyres	Values of tightening force, minimum and maximum, to be mentioned on the drawing.	WDK 116	F0	 (070)	Supplier

5.3.2.3 FC13: The tyre shall be compatible with the lubricant used."

Characterization of elements of the external environment of use: see FP6.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Compatibility in the lubricant for assembly 4 tests mini + industrial tests	No rupture, crack or swelling must be observed after the test	Fittability test in Plant	F0		Supplier Renault / Nissan

5.3.2.4 FC14: The tyre has to respect its technical environment.

Characterization of elements of the external environment of use: see FP6.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Closeness of the rim protector of wheel rim. Grading of the roll of protection according to functional requirements.	The shape and size of the rim protector shall not hamper the assembly and the holding of the balancing weights assembled on wheel rim. The tests of balancing must be considered satisfactory by the assembly plant. Geometry of rim protector to be described on drawing (see Annex F)	Balancing test in Plant	F0		Supplier Renault/ Nissan

5.3.2.5 FC15: The tyre must inform the assembly operator for mounting orientations.

Characterization of elements of the external environment of use: see FP6.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Insure the orientation fitment of the tyre : 1) Non directional Tyre 2) Directional Tyre Markings of the tyre to be stated and identified on the drawing.	Marking " Outside " or a direction of rotation. Presence of a red dot to help the operator to assemble the tyre in the appropriate orientation. Definition of Red Dots to be considered as Annex I	1) Non directional tyre : Marking of a red point on the outside side of the tyre specifying the maximal value of 1st harmonic of Variation of Radial force (RFV H1). 2) Directional tyres : Presence of a detail indicating clearly the direction of rotation of the tyre. Marking of a different colour point on the outside side for the tyre of the right and on the internal side for the left tyre. Both dots have to identify RFVH1 Max position (mandatory in case of Matching)	F1 F0 F1		Supplier
Marking Lines	Supplier must indicate on drawing definition (used color sequence) and position of marking lines used to identify the tyre reference	Drawing review	F2		Supplier

5.3.3 LIFE PHASE "Tyre inflation"

5.3.3.0 UCEps : Unwanted Customer Events potentially safety related

The supplier must take into account the list of Events potentially safety below, relevant for the tire. These FCEps are extracted from the common Alliance list.

- FECps n°18: Loss of control: due to a sudden loss of the pressure or the explosion of a tire (except aggression by elements outside the vehicle). (during the driving)

The supplier is in charge to:

- complete the safety events list based on its experience, expertise or analyses.,
- give to Renault or Nissan the details linked to these events and the use of his product on vehicle,
- to be able to attest how the safety events have been taken into account in product design and manufacturing process of the tire. (Design Failure Mode and Effects Analysis – see ANPQP requirements)
- provide demonstration elements (refer §7 RELIABILITY – DURABILITY)

5.3.3.1 FP7 : The tyre shall allow pressurization and the adjustment of pressure by the inflation system.

Characterization of the elements of the external environment of use: (additional to FP1):

INFLATION DEVICE

In Plant : Inflating bell (Pressure 3 bars or 3,5 bars according to ETRTO recommendation).

In After-sales : Inflating Gun ex: Schrader R 1822-1 (12 maxi bars) or workstation of inflation ex: EMIC SG 1000 (6 maxi bars).

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Pressurisation of the tyre 4 tests mini + industrial tests.	The supplier will have to guarantee the possibility of tyre inflation with the equipment of the after-sales network. For Passenger car tyres : The assembly must be considered satisfactory for Maximum pressure of 3.5 bar (350Kpa) For commercial vehicle tyres : The assembly must be considered for Maximum pressure of 4.5 bar (450Kpa) when standard valves are used and 5.75bar (575Kpa) when TR600 valves are used. The supplier will have to indicate the maxi pressure of assembly of his tyre.	None	F0		Supplier Renault / Nissan

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Inflating bell compliance	Tire design should not affect plant inflating device performance (e.g. marking position)	Test of inflating in factory	F0		Supplier
Pressurisation of the tyre 4 tests mini + industrial tests.	The tests of inflating must be considered satisfactory by the assembly plant for all the rims of the project	Test of inflating in factory	F0		Supplier Renault / Nissan

5.4 "FUNCTIONS/LIFE PHASES" SUMMARY TABLE

	User	Assembly	Inflating
FP 01 "Must bear vehicle load transmitted by rim"	X		
FP 02 "Transmitting forces between rim and ground"	X		
FP 03 "Transmit static electricity from the rim to the ground"	X		
FP 04 "Filtration of forces by the ground the rim"	X		
FP 05 "ParticipateEnergy performance of the vehicle"	X		
FP 06 "Must be installed on the wheel with mounting device"		X	
FP 07 "Allow pressurization and adjustment of pressure"			X
FC 01 "Must comply with the used "	X		
FC 02 "Must give information with the user"	X		
FC 03 "Must respect the people outside the vehicle"	X		
FC 04 "Must resist the ground"	X		
FC 05 "Must resist the external environment"	X		
FC 06 "Must respect its technical environment"	X		
FC 07 "Must resist the inflating gas"	X		
FC 08 "Must resist the repairing product"	X		
FC 09 "Must respect external environment"	X		
FC 11 "Must comply with the regulations in force"	X		
FC 12 "Must be installed on the rim"		X	
FC 13 "Must be compatible with the lubricant used"		X	
FC 14 "Must comply with its technical environment"		X	
FC 15 "Must inform its fitter about its orientation of installation"		X	
FC 16 "Must comply with the fitter"		X	

6. FUNCTIONAL DEFINITION for Temporary Spare tire only

6.1. PRODUCT LIFE CYCLE

LIFE PHASES	studied	Not studied
LOGISTICS – Packaging – Storage 1 – Transportation – Acceptance – Storage 2 – Unpackaging – Delivery rate		X X X X X X X
ASSEMBLY – Preparation before assembly – Assembly on vehicle – Startup in plant and inspections	X X X	
DELIVERY OF THE VEHICLE – Vehicle storage – Vehicle transportation – Vehicle placed at disposal		X X X
UTILISATION BY FINAL CUSTOMER – stopped – running – braking – subject to impacts – parking – cleaning (brushes, high pressure jet, manual cleaning)	X X X X X X	

6.1.1. CHARACTERIZATION OF FUNCTIONS (verb + element describing the external environment-EME)

Explanations:

Criterion : figures or qualities to define the performance expected from the product.

Level : rank on criteria measurement scale.

Test Method: method allowing to check that the level has been reached.

Flexibility (indicates whether a criterion can be negotiated or whether it is mandatory) :

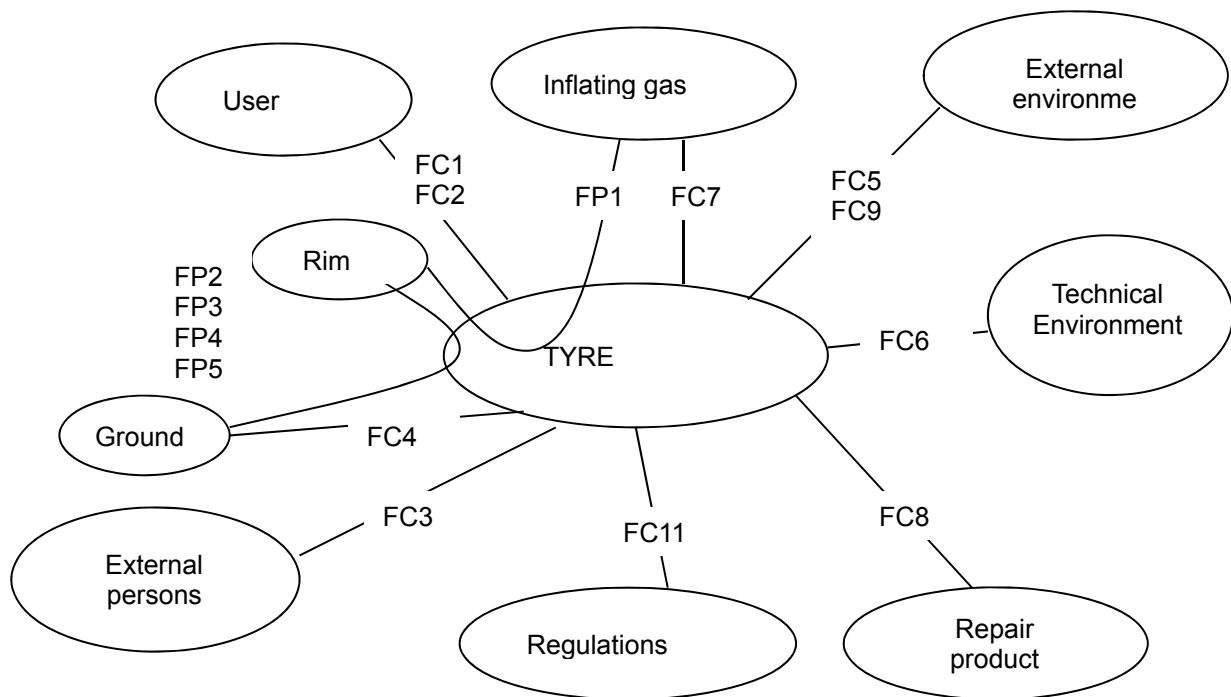
- F0 : no flexibility, mandatory ,
- F1 : low flexibility, less negotiable.
- F2 : good flexibility, negotiable,
- F3 : very good flexibility, very negotiable.

HCPP : Product /Process Characteristic Hierarchy. The Rate of Non Compliance (TNC) associated with each severity class are indicated in the Table of Disturbed Service Functions (FSP) figuring in [Annex D](#)

Person in charge : person in charge of checking that the level has been respected.

6.2 LIFE CYCLES:

6.2.1 Life cycle phase: "Using by the Customer"



Functions to be ensured

FP : Main Function and **FC**: Constraint Function

- FP1: The tyre must bear the vehicle load transmitted by the rim, supported by the inflating gas.
- FP2: The tyre shall transmit the forces between the rim & ground (acceleration, braking, cornering, etc.).
- FP3: The tyre must transmit static electricity from the rim to the ground.
- FP4: The tyre must filter the forces transmitted by the ground to the rim (passenger comfort, and preserve the integrity of the parts stressed by it).
- FP5: The tyre participates in the energy performance of the vehicle.
- FC1: The tyre must comply with the user (running noise heard by the occupants of the vehicle from inside the vehicle, ...).
- FC2: The tyre must give information to the user (marking, wear indicator, certification, etc.).
- FC3: The tyre must respect the people outside the vehicle (external noise, appearance, etc...).
- FC4: The tyre must resist the ground (perforation, heat, soil type, etc..) .
- FC5: The tyre must resist the external environment (washing product, hydrocarbons, etc.).
- FC6: The tyre must respect its technical environment (body, axle, wheel cover, speedometer, ABS system...).
- FC7: The tyre must resist the inflating gas.
- FC8: The tyre must resist the repairing product.
- FC9: The tyre must respect the external environment (non-polluting material, ...).

- FC11: The tyre must comply with the regulations in force.

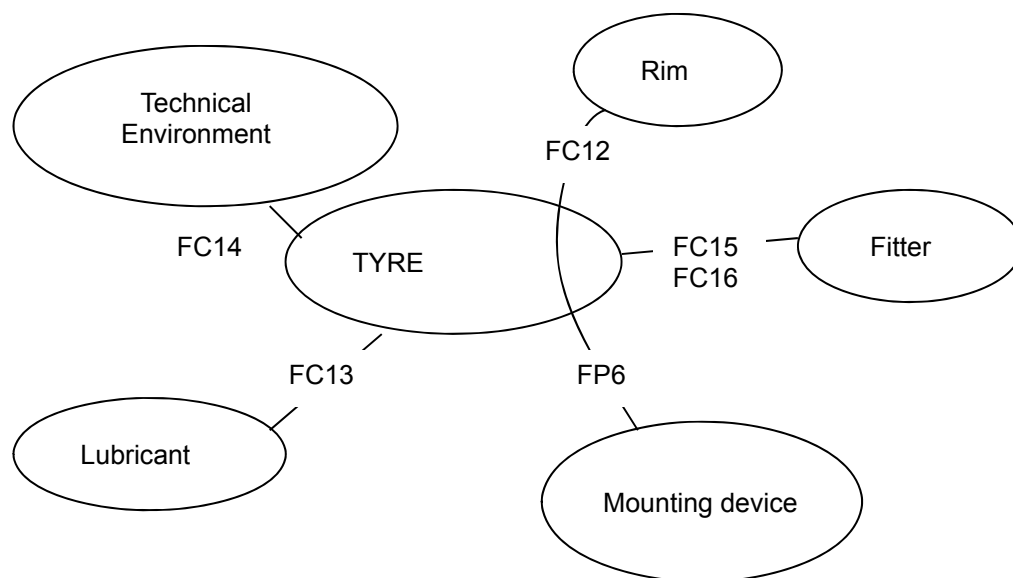
6.2.2 Life cycle phase: "ASSEMBLY"

Functions to be ensured

- FP6: The tyre must be installed on the wheel with the mounting device

- FC12:
must be
on the

- FC13:
shall be



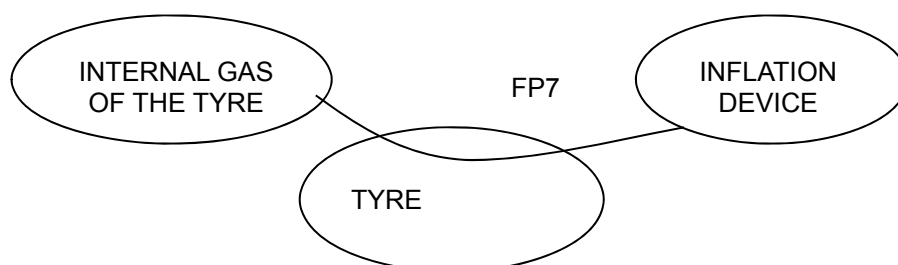
The tyre
installed
rim.

The tyre

compatible with the lubricant used.

- FC14: The tyre must comply with its technical environment (balancing weight, roller bench, etc.).
- FC15: The tyre must inform the fitter about its direction/orientation of installation.
- FC16: The tyre must comply with the fitter.

6.2.3 Life cycle phase: "Tyre inflation"



Functions to be ensured

- FP7 : The tyre shall allow pressurization and the adjustment of pressure by the inflation system.

6.3 CHARACTERISATION OF FUNCTIONS

6.3.1 Life cycle phase: "Use by the Customer"

6.3.1.0 UCEps : Unwanted Customer Events potentially safety related

The supplier must take into account the list of Events potentially safety below, relevant for the tire. These FCEps are extracted from the common Renault Nissan list.

The supplier must take into account the list of Events potentially safety below, relevant for the tire. These FCEps are extracted from the common Alliance list.

- FECps n°11: Loss of control: due to an absence of braking or an unexpected loss of the braking or a loss of assistance of braking not indicated, including due to a driving with parking brake tightened not indicated. (during the driving)
- FECps n°18: Loss of control: due to a sudden loss of the pressure or the explosion of a tire (except aggression by elements outside the vehicle). (during the driving)

The supplier is in charge to:

- complete the safety events list based on its experience, expertise or analyses.,
- give to Renault or Nissan the details linked to these events and the use of his product on vehicle,
- to be able to attest how the safety events have been taken into account in product design and manufacturing process of the tire. (Design Failure Mode and Effects Analysis – see ANPQP requirements)
- provide demonstration elements (refer §7 RELIABILITY – DURABILITY)

6.3.1.1 FP1 : The tyre must bear the vehicle load transmitted by the rim, supported by the inflating gas.

Characterization of the elements of the external environment of use:

User: vehicle driver

Rim : Rim made of steel sheet or stamped aluminium which is coated of protection (Electrostatic painting or paint)

Light alloy rim moulded or forged which is coated with protection (paint + varnish).

The definition of the tyre and wheel interface is compliant with ETRTO.

The dimensions of the rims to be used for testing are indicated in the functional file of the tyre

The rim used for the tests will always be the one intended for the vehicle. Its definition can change over time depending on the progress of the vehicle in the project. The rim will be provided by Renault-Nissan, if not available, a rim of similar size in terms of width and wheel offset must be used in accordance with Renault-Nissan. If some tests require the use of different size of rim, it must be stated in the test report with the results.

The minimum and maximum inflating pressure at ambient temperature If the supplier has recommendations different from those described in ETRTO, he must inform about them.

Reference pressure (inflation pressure for testing): Unless otherwise indicated, use the ETRTO reference pressure indicated by "INFLATION PRESSURE" in the tables of the series "METRIC DESIGNATION" of tyres.

Maximum speed: The design speed of the temporary-use spare unit shall be at least 120 km/h for types 1, 2 and 3.

Maximum loads of the vehicle that the front and rear axles can bear: The tyre delivered by the manufacturer must have a load index compatible with the loads of the vehicle on which it will be mounted.

Internal gas : pressurised ambient air, nitrogen, helium (possible in after-sales service).

Technical Environment

Wheel arch, elements of axles (suspension etc.), wheel cover, balancing weight, compartment of spare wheel (inside the vehicle in the trunk, outside under the body) in a wire carrier or maintained by a winch, or fixed to the body outside the vehicle).

Additional attachment device:

The additional attachment device must be in compliance with the French and European standards. The size of the chain planned on the vehicle is mentioned in the functional file of the tyre.

Repairing product: Inflating kit (reference indicated in the functional file of the tyre).

Environment:

- Temperatures: between - 40 °C and 60 °C (except for prescriptions mentioned in the functional file of the tyre).
- Fluids: hydrocarbons, snow clearing solution, brake fluid, washing fluids, liquid from the tyre repair kit.
- Ozone.
- Gravel, dust, water, mud, snow, etc.

External environment: Ground more or less regular (asphalt road, concrete road, paved road, dirt tracks, etc.), pavements, etc.

Characterization of the assessment criteria:

Details on test conditions:

Subject to the compliance by the manufacturer with the tyre dimension (including load and speed indices), the assessment of the tyre in the function FP1 is covered by the other main functions and constraints described in this specification.

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Sealing at room temperature	Loss of pressure less than 0.1 bar after 1 month of storage	6 tyres assembled on wheel rim Initial pressure = ETRTO reference pressure. Continuous checking of the pressure without disconnecting the pressure gauge from the valve T = 23 °C ± 5°C.	F0		Supplier
Load/Speed performance test	The tyre must have no degradation (separation or pulling of the tread, cracking of folds, etc.).	Minimum 1 mini tyre mounted on the rim of vehicle (1) E / ECE / 30 for Passenger car E / ECE / 54 for Commercial Vehicles (ECE R30/R54 marking is required for all destination)	F0		Supplier
Speed index	The design speed of the temporary-use spare unit shall be at least 120 km/h for types 1, 2 and 3.	Compliant with E / ECE / 64 for tires types 1, 2 and 3.	F0		Supplier
Tubes specifications of tyres with tube	Tubes specifications shall comply with JIS D4207	Supplier Drawing	F2		Supplier

(1): Test necessary for the certification of tyre, to be conducted on the final fabrication selected during various tests & declared as such by Renault / Nissan

6.3.1.2 FP2 : The tyre must transmit the forces between the wheel rim and the ground

FP4: THE tyre **must filter the forces transmitted** by the ground to the wheel rim

Characterization of elements of the external environment of use: see FP1.

The supplier must be able send an F-Tire model of the tyre defined in the functional file in case of request expressed by Renault. This is used to create simulations of driving on endurance track. See [Annex A](#)

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Comfort, Ride and Handling	vehicle stability Validation equipped with a spare assembly at each wheels position (with other tires brand /sizes)	1 spare tyre mounted on the vehicle. Subjective evaluation and verification for the tire at each position on vehicle. Applicable test procedures : 3108882 & Nissan Method	F1		RENAULT / NISSAN
Bead unseating test	Value of force according to the load Index of the tyre, more than:	Minimum 2 tyres on the test bench. ISO 10191	F0		Supplier
	For $LI \leq 75 \Rightarrow F > 6\,670\text{ N}$ For $76 < LI \leq 92 \Rightarrow F > 8\,890\text{ N}$ For $LI \geq 93 \Rightarrow F > 11\,120\text{ N}$	Test on the wheel rim of the project			

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Braking and deviation test for vehicles fitted with temporary use spare units	The stopping distance achieved using a maximum force of 500 N applied to the foot control shall not exceed 50.7 m and; The mean fully developed deceleration (mfdd) given by the following formula shall be not less than 5.8 ms⁻²: $Mfdd = v^2 / 41.14 \text{ s}$ where "v" is the initial speed at which braking commences and "s" is the distance covered during braking between 0.8 v and 0.1 v.	ECE 64 Annex 3 In the case of M1 category vehicles approved to Regulation No. 13 fitted with temporary-use spare units and tested using a prescribed speed of 80 km/h.	F0		Supplier Renault / Nissan
Static unbalance	Below 0.118 N.m	ISO 13326	F1		Supplier

6.3.1.3 FP3 : The tyre must transmit static electricity from the rim to the ground.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Electrical Resistivity Electric resistance between Tire Bead and Tread Center	Resistivity of the tyre must be less than 108 Ohms	WDK 110 Minimum 3 tyres mounted on the steel wheel rim of the vehicle (If they exist).	F0		Supplier

6.3.1.4 FC1: The tyre must comply with the user.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Visual appearance of the tyre in delivery conditions	The supplier will have to present, tyres sorted and representative of what he considers as manufacturing acceptance limits for the dimension and the brand line which he proposes V1 - Strong criticism Indentation upper to 1.4mm V2 - Potentially open to criticism 0.9 mm < Indentation <=1.4 mm V3 - Non critical : Indentation <0.9 mm The concave on the side wall of the tire shall be 0.8 mm @400 kPa or less as a guideline. The tire should not show any visible texture due to curing/reworking.	A check of the general appearance will be made and validated by the representatives of the perceived Quality, in particular on the possible indentations of the sidewall of tyres.	F3		Supplier Renault / Nissan

The supplier will have to send a Finite Element Model of the tyre selected during the various tests and declared as such by Renault. The FEA model will have to comply with specifications provided by Renault/Nissan.

6.3.1.5 FC2: The tyre must give information to the user (marking, wear indicator, certification, etc.).

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Marking Dimension and characteristics of the tyre (speed and load indexes, direction of rotation, type of structure, use, etc.	The designation must be in compliance with the markings required by the regulations. Definitions and positions to be indicated on the drawing. ECE30/54 regulation requirements (markings and certificates) must be applied for all Tyres independently of commercialization countries.	Regulations : See the project functional file to identify the regulations defined according to the geographical zones : Example: 92/23/CE Or E/ECE/117 E / ECE / 30 for passenger cars E / ECE / 54 for commercial vehicles	F0	 (070)	Supplier
Wear indicator	Minimum 6 indicators per groove with an identification mark on the sidewall of the tyre. Definitions and positions to be also indicated on the drawing. Height of indicator: 1.6 mm +0.6 mm/-0mm;(for summer tires)	control report on 1 tyre	F0	 (070)	Supplier

6.3.1.6 FC4: The tyre must resist the ground.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Wear	3500 km or more under normal operation before exposure of “tread wear indicator”.	Vehicle driving test according to supplier test specification. Tire fitted to vehicle drive axle position Inflation pressure: 420 kPa	F1		Supplier
Durability	Run continuously for 750 km or more without stopping	Bench test Load: ETRTO Max load Speed: 100km/h Inflation pressure: 420 kPa	F0		Supplier

6.3.1.7 FC5: The tyre has to resist the external environment.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Initial properties in the tension Minimum 4 samples per type of material composing the side wall and the tread of a tyre.	Tensile test in an initial state Resistance to rupture Rr (in MPa) Elongation at rupture Ar(in %) No requirement, Rr and Ar are the references to determine the variations after ageing.	D41 1099	F1		Supplier
Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge

Resistance to hydrocarbons (1) Minimum 4 samples per type of material composing the side wall and the tread of a tyre	Immersed in the diesel fuel: 15s 23°C, then storage 24 @ 23°C. Immersed in petrol: 15s 23°C, then storage 24 @ 23°C. No rupture, crack or swelling must be observed Tensile test on old samples (immersion + storage) Variation R_r / initial = 20 % Variation A_r / initial = 30 %	D471924 Fluids O and Y according to standard 03-50-000 D41 1099	F1		Supplier
Resistance to brake fluid (1) Minimum 4 samples per type of material composing the side wall and the tread of a tyre.	Immersed in the fluid for 15s 23°C, then stored 24 hours @23°C. No rupture, crack or swelling must be observed after the test	D471924 Fluid type 3 according to standard 41-02-001 D41 1099	F1		Supplier
Resistance to zinc chloride (1) Minimum 4 samples per type of material composing the side wall and the tread of a tyre.	Immersed in the fluid (Fluid constituted in weight by 50 % of chloride and the rest water) for 24 hours 23°C, then stored 24 hours @23°C. No rupture, crack or swelling must be observed after the test Tensile test on old samples (immersion + storage) Variation R_r / initial = 20 % Variation A_r / initial = 30 %	D471924 D41 1099	F1		Supplier

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Resistance to calcium chloride (1) Minimum 4 samples per type of material composing the side wall and the tread of a tyre.	Immersed in the fluid (Fluid constituted in weight by 50 % of chloride and the rest water) for 24 hours 23°C, then stored 24 hours @23°C. No rupture, crack or swelling must be observed after the test Tensile test on old samples (immersion + storage) Variation R_r / initial = 20 % Variation A_r / initial = 30 %	D471924 D41 1099	F1		Supplier
Heat resistance (1) Minimum 4 samples per type of material composing the side wall and the tread of a tyre.	After exposure for 7 days @ 70°C and return to 23°C. Tensile test on old samples Variation R_r / initial = 20 % Variation A_r / initial = 30 %	D47 1053 D41 1099	F1		Supplier
Resistance to ozone (1) Minimum 4 samples per type of material composing the side wall and the tread of a tyre.	Concentration of 200 ppcm during 24 hours 40°C. No cracking or fissures observed after test	D47 1100	F1		Supplier

(1) If a different procedure exists at the supplier's, its equivalence must be demonstrated. After approval from Renault, the supplier can use it. The validation is made in a generic way for the used materials and not for every tyre.

6.3.1.8 FC6: The tyre must comply with its technical environment.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Circumference of tyre. Tyre mounted on the rim of the vehicle.	The nominal value of the rolling circumference of the tyre has to be in accordance with the ETRTO. The tolerance has to be in compliance with the WDK 128 (+1,5% -2, 5%). (1):	WDK 107	F1	 (048; 054)	Supplier
Static loaded Radius Tyre mounted on the rim of the vehicle.	The value of the static loaded radius of the tyre has to be in accordance with the tolerance of WDK 128 (± 2 %). (2):	WDK 107	F1		Supplier

- (1) It is also requested to measure the rolling circumference in the pressures and the load defined to the functional document of the tyre according to the speed (from 20 km/h to maximal speed of the tyre).
- (2) It is also requested to measure the radius loaded to the pressures and the load defined to the functional folder of the tyre according to the load (from 0 kg to the max load of the tyre).

6.3.1.9 FC7: The tyre must resist the inflating gas.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Behaviour to the nitrogen / Helium Sealing at ambient temperature 2 tyres assembled on wheel rim.	Loss of pressure lower than 0,1 bar after 1 month of storage. The supplier has to indicate any incompatibility with these gases.	Initial pressure Benchmark pressure of ETRTO Continuous control of the pressure without disconnecting the manometer of the valve. T = 23 °C ± 5°C	F1		Supplier

6.3.1.10 FC8: The tyre must resist the repairing product.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Resistance to repairing fluid : 2 tyres assembled on wheel rim	In the recommended practices of use, the supplier will have to indicate if it is the case, the limitations of use of product of repair on its tyres.	None	F2		Supplier

6.3.1.11 FC9: The tyre has to respect the outer environment.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Component materials the tyre. completion of the material data sheet IMDS (mandatory for RENAULT and on request for NISSAN).	No substances with prohibited use No aromatic hydrocarbons.	Standard 00 10 050 Guideline 2005 / 69 / CE (Guideline 2000 / 53 / CE) REACH	F0		Supplier
Recycling	The manufacturer will be organized to manage the worn tyres and participate actively with the collection and valuation sectors of the worn tyres.	Without	F0		Supplier
Weight	The mass has to be in accordance with the objective value of the Spec. Tender / Functional file of the tyre +/-3 %.	Capability must be demonstrated according to requirement of the ability contract	F1		Supplier

6.3.1.12 FC11: The tyre must comply with the regulations in force.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
The supplier must comply to perform all that would be necessary and tests under its responsibility in order to obtain the necessary approval for the industrialization of its tyre then for its use in these various countries. The supplier will have to provide certification numbers of his product and supply the certificates at the request of RENAULT/NISSAN Commercialization countries to be considered are included in Spec. Tender / Functional files of the project.	The tyre has to be in accordance with all the set of all criteria mentioned in the regulations and/or the guidelines. Information to be shown on the supplier drawing according to the format of the standard drawing.	According to regulatory requirements	F0	 (070)	Supplier

6.3.2 LIFE PHASE “Assembly”

6.3.2.0 UCEps : Unwanted Customer Events potentially safety related

The supplier must take into account the list of Events potentially safety below, relevant for the tire. These FCEps are extracted from the common Alliance list.

- FECps n°18: Loss of control: due to a sudden loss of the pressure or the explosion of a tire (except aggression by elements outside the vehicle). (during the driving)

The supplier is in charge to:

- complete the safety events list based on its experience, expertise or analyses.,
- give to Renault or Nissan the details linked to these events and the use of his product on vehicle,
- to be able to attest how the safety events have been taken into account in product design and manufacturing process of the tire. (Design Failure Mode and Effects Analysis – see ANPQP requirements)
- provide demonstration elements (refer §7 RELIABILITY – DURABILITY)

6.3.2.1 FP6 : The tyre must be set up on the wheel with the mounting device

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Assembly of the tyre on wheel rims. 4 tests mini + industrial tests.	The assembly must be considered satisfactory by the assembly plant on all the wheel rims of the project..	Fittability test in Plant	F0		Supplier Renault / Nissan

6.3.2.2 FC12: The tyre must be installed on the rim.

Characterization of elements of the external environment of use: see FP1.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Bead base clamping Force. Test of measures of effort on 6 minimum tyres	Values of tightening force, minimum and maximum, to be mentioned on the drawing.	WDK 116	F0		Supplier

6.3.2.3 FC13: The tyre shall be compatible with the lubricant used."

Characterization of elements of the external environment of use: see FP6.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Compatibility in the lubricant for assembly 4 tests mini + industrial tests	No rupture, crack or swelling must be observed after the test	Fittability test in Plant	F0		Supplier Renault / Nissan

6.3.2.4 FC15: The tyre must inform the assembly operator for mounting orientations.

Characterization of elements of the external environment of use: see FP6.

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Marking Lines	Supplier must indicate on drawing definition (used color sequence) and position of marking lines used to identify the tyre reference	Drawing review	F2		Supplier

6.3.3 LIFE PHASE "Tyre inflation"

6.3.3.0 UCEps : Unwanted Customer Events potentially safety related

The supplier must take into account the list of Events potentially safety below, relevant for the tire. These FCEps are extracted from the common Alliance list.

- FECps n°18: Loss of control: due to a sudden loss of the pressure or the explosion of a tire (except aggression by elements outside the vehicle). (during the driving)

The supplier is in charge to:

- complete the safety events list based on its experience, expertise or analyses.,
- give to Renault or Nissan the details linked to these events and the use of his product on vehicle,
- to be able to attest how the safety events have been taken into account in product design and manufacturing process of the tire. (Design Failure Mode and Effects Analysis – see ANPQP requirements)
- provide demonstration elements (refer §7 RELIABILITY – DURABILITY)

6.3.3.1 FP7 : The tyre shall allow pressurization and the adjustment of pressure by the inflation system.

Characterization of the elements of the external environment of use: (additional to FP1):

INFLATION DEVICE

In Plant : Inflating bell (Pressure 4.2 bars to 4.5 bars according to ETRTO recommendation).

In After-sales : Inflating Gun ex: Schrader R 1822-1 (12 maxi bars) or workstation of inflation ex: EMIC SG 1000 (6 maxi bars).

Characterization of the assessment criteria:

Criterion	Level (value +tolerance range+ unit)	Procedure / Test Method	Flexibility	HCPP	Person in charge
Pressurisation of the tyre 4 tests mini + industrial tests.	The supplier will have to guarantee the possibility of tyre inflation with the equipment of the after-sales network. The assembly must be considered for a pressure of 4.2 bar (420Kpa). The supplier will have to indicate the maxi pressure of assembly of his tyre.	None	F0		Supplier Renault / Nissan
Inflating bell compliance	Tire design should not affect plant inflating device performance (e.g. marking position)	Test of inflating in factory	F0		Supplier
Pressurisation of the tyre 4 tests mini + industrial tests.	The tests of inflating must be considered satisfactory by the assembly plant for all the rims of the project	Test of inflating in factory	F0		Supplier Renault / Nissan

6.4 "FUNCTIONS/LIFE PHASES" SUMMARY TABLE

	User	Assembly	Inflating
FP 01 "Must bear vehicle load transmitted by rim"	X		
FP 02 "Transmitting forces between rim and ground"	X		
FP 03 "Transmit static electricity from the rim to the ground"	X		
FP 04 "Filteration of forces by the ground the rim"	X		
FP 05 "ParticipateEnergy performance of the vehicle"	X		
FP 06 "Must be installed on the wheel with mounting device"		X	
FP 07 "Allow pressurization and adjustment of pressure"			X
FC 01 "Must comply with the used "	X		
FC 02 "Must give information with the user"	X		
FC 03 "Must respect the people outside the vehicle"	X		
FC 04 "Must resist the ground"	X		
FC 05 "Must resist the external environment"	X		
FC 06 "Must respect its technical environment"	X		
FC 07 "Must resist the inflating gas"	X		
FC 08 "Must resist the repairing product"	X		
FC 09 "Must respect external environment"	X		
FC 11 "Must comply with the regulations in force"	X		
FC 12 "Must be installed on the rim"		X	
FC 13 "Must be compatible with the lubricant used"		X	
FC 14 "Must comply with its technical environment"		X	
FC 15 "Must inform its fitter about its orientation of installation"		X	
FC 16 "Must comply with the fitter"		X	

7. RELIABILITY - DURABILITY

Supplier is responsible to commit :

- to the quality and the reliability of the supplied part (design, manufacturing –Product Quality and Reliability Targets Study - see ANPQP requirements).
- To inform RENAULT/NISSAN of the limits of use and the utilization limits of the supplied product (Supplier Recommendations – see ANPQP requirement).

8. DEPENDABILITY

General information:

Possible malfunctions of products are dealt with through a dependability-type procedure in order to meet Renault and Nissan requirements in terms of reliability, availability and safety.

This procedure must be adapted so that it fits the specificities of the product to be developed.

This explains why the Supplier must meet the requirements defined according to the procedure hereafter.

8.1. WORK PROGRAMME

8.1.1 Step 0: using Renault and Nissan documents

The Supplier uses the documents drafted by RENAULT AND NISSAN (Product Specifications, etc.).

The Supplier refers to the functional analysis, drafts block-diagrams and the packaging diagram which are prerequisites for clearance of risks.

8.1.2 Step 1: Identification of Customer Undesirable Events (EIC)

As soon as the pre-contract is signed, the Supplier takes part in system studies through work groups (PFE) set up by RENAULT AND NISSAN.

The activities include drafting and updating of undesirable events and the definition of the tools (FMECA, failure trees or any other tool) to use for bringing to the fore the risks associated to each one of the undesirable events. Besides, these tools must be applied to the system.

These documents must be furnished by RENAULT AND NISSAN PFE (New Vehicle Function Team) as regards the contract and they are updated later.

The undesirable events that have been identified up to day and needing to be dealt with (failure trees and/or FMECA) must be included on the Quality Single List (LUQ) . This concerns the components for which an experience feedback is required. As regards product and/or process innovations, a specific list and handling procedure must be created.

EIC number	EIC	max. occurrence rate
	(to be completed)	

This list of Customer Undesirable Events can be completed while studying dependability according to test results and prototypes, etc. The Provisional Analysis of Risks is one of the information sources.

8.1.3 Step 2: Dealing with Undesirable Events

Specific studies are carried out by the Supplier on products in parallel to RENAULT AND NISSAN studies. As regards Product and/or Process innovations, these studies include:

a Technical Functional Analysis,

a failure tree or several detailed failure trees,

handling of quality and quantity matters appearing on these failure trees (possible occurrence of customer undesirable events)

a Provisional Analysis of Risks,

FMECA (Failure Modes, their Effects and Criticality Analysis) relative to components (product/process) when the solutions are innovative and/or when the risks are identified,

provisional reliability studies of the system,

any action considered as appropriate by the Supplier.

At this stage, the Supplier is totally responsible and progress meetings specific to Dependability can be organized at RENAULT AND NISSAN' s or at the Supplier's request or within the framework of project progress meetings. RENAULT AND NISSAN reserves the possibility to carry out a diagnosis of analysis contents and of relevance. The Supplier shall communicate to RENAULT AND NISSAN the analyses necessary to prove that quality is guaranteed.

8.1.4 Step 3: action plan

Action plans are carried out by the Supplier and/or Renault and Nissan. Monitoring is part of the Project Progress meetings. These meetings shall take place once a month at least. The subjects to be dealt with in this document are those mentioned in the table below :

Type of problem identified	Action to be taken	Person in charge	Deadline	Result and progress

8.2. INFORMATION TO BE PROVIDED

- the Supplier shall participate in the various meetings organized by the « dependability » work group,
- the Supplier shall participate in the various meetings organized by the HCPP work group,
- FMECA system (*),
- FMECAs concerning components and summary (product/process) (*),
- Failure trees and handling processes (*),
- Quantified evaluations of provisional reliability (*),
- Action plan,
- Action plan progress,
- Possible diagnosis from RENAULT AND NISSAN based on the analysis documents furnished or participation of RENAULT AND NISSAN in analysis groups.
- Supplier shall provide Control Report which shows the tire performance equivalency between spec frozen tyre and each physical lot supply tyre, before starting each physical lot by using static/dynamic characteristic data comparison and/or vehicle evaluation result.

(*) Decision made by the work group when studying the system, if need be.

8.3. Provisional Analysis File

The Provisional Analysis File falls within the scope of the upstream ANPQP procedure (Alliance New Product Quality Procedure) created by RENAULT AND NISSAN Purchasing Department. Among others, the Supplier shall furnish a reliability provisional analysis file (FMECA, failure trees, provisional calculations from database, etc.).

8.4. Validation procedure

Before starting mass-production and before accepting any modification, product's reliability must be evaluated.

The inspection procedure is based on the objectives set for the project in question.

The following requirements are jointly defined by the Supplier, RENAULT AND NISSAN PFE, the Purchasing Department and the Quality Department:

- The number of parts to be tested,
- The test(s) to be performed,
- The test conditions (see table to be completed by the PFE),
- The reliability level (representativeness).

9. QUALITY

Quality objectives must make it possible for RENAULT/NISSAN to meet the expectations of its customers during the warranty period of the vehicle. The facilities that Suppliers use to meet these objectives must make it possible for Renault to assess their performance.

If relevant, targets are included in project functional documents.

Part :		
GMF 3 months of Use		K°/oo
PIMOF 3 months of Use		K°/oo
GMF 12 months of Use		K°/oo
PIMOF 12 months of Use		K°/oo
GMF 24 months of Use		K°/oo
PIMOF 24 months of Use		K°/oo
GMF 36 months of Use		K°/oo
PIMOF 36 months of Use		K°/oo

GMF: Manufacturing Month Warranty

PIMOF: Off-road Breakdown per Manufacturing Month

10. CONSTRAINTS NOT LINKED TO THE USE

10.1 SUPPLIER'S RESPONSIBILITY

The provisions of the standard 00-10-415 "Production and supply of products – general requirements" are fully applied to the present specification, except "the marking of products" and other express and written contrary provisions.

10.2 SUPPLIER DRAWING

The various sheets which make up the Supplier drawing will have to be in accordance with the standards drawings provided by Renault / Nissan at their last revision level. The basic informations (dimensions, markings, pattern, Renault first sheet content...) is to be supplied at the end of the JT #1 or initial submission and the full set of informations (definitive data, dynamic characteristics...) is to be supplied at the end of the JT #3 or final submission. Items which could not be informed or certified to these dates must be indicated as reserves in a dedicated chart on drawing ("Table of reserves"). This one must be cancelled during the definitive specification of the plan. Differences versus characteristics required during the RFQ will be grouped in a deviation table.

10.3 PACKAGING

No special condition is required.

10.4 AVAILABILITY OF TYRES IN AFTER-SALES

The particular constraints bound to the after-sales are based on the following points:

- Availability of spare tyres in all the countries where is sold the vehicle, from the marketing, and it during the life expectancy of the vehicle. The dimensions (basis, option, surmounts and spare wheel) must be available at least 10 years after the output of the last vehicle of the factory.
- Manufacturer logistics must be able to deliver just in time to vehicle dealers.
- For every dimension retained and sold in a country susceptible to be subjected to a cold period (winter, snow), the supplier will have to propose aftersales solutions of winter and studded tyre. The constraint of 10 years of availability also applies to these type of tyre.



10.5 EXPIRY OF TYRES ^R(070)

Tyres delivered to the tyre assembly plant must not present of DOT more than 52 weeks on reception in these plants. This requirement must be reduce to 6 months in case of specific requirement,for example, related to specific country regulation.

Annex A - FTYRE

(Normative)

The validation durability of vehicles leans on notes of measure on "track" "H6-12-2" "" of Lardy. This track includes several types of obstacles including potholes, speed bumps, bad pavements etc.

The vehicle is equipped with dynamometric wheels, accelerometers and stress gauges. After acquisitions on circuit, the vehicle is assembled on a bench of test (Bench 12 power cylinder for example) reproducing the requests seen on the circuit.

To reproduce this process via the digital sector (ADAMS calculation), it is necessary to build a model of complete vehicle and to make it run on the virtual circuit (digitalization of the real circuit).

The purpose is to obtain the dimensioning requests on the parts of suspension and the box. This methodology of simulation requires a model of appropriate tyre. The Renault computation department uses the model F-tyre.

To provide

For a size of tyre (choice makes by RENAULT):

A.1 Input data F-car in the standard .tir format including the characterization for 2 pressures of different tyre

According to the example of .tir dataset (example of standard supplied with the license

Ftyre / ADAM)

\$-----MDI_HEADER

[MDI_HEADER]

FILE_TYPE = 'tir'

FILE_VERSION = 3.0

FILE_FORMAT = 'ASCII'

(COMMENTS)

{comment_string}

'Tyre Manufacturer - xxxxxxxxxxxxxx'

'Tyre Type - xxxxxxxxxxxxxx'

'Tyre Dimension - 195/65 R 15'

'Pressure - 2.0 bar'

'File Generation Date - xx/xx/xx xx:xx'

\$-----UNITS

[UNITS]

FORCE = 'NEWTON'

MASS = 'GRAM'

LENGTH = 'MM'

TIME = 'MILLISECOND'

ANGLE = 'DEGREE'

\$-----DIMENSION

[DIMENSION]

UNLOADED_RADIUS = 326.0 \$ [mm]

\$-----VERTICAL

[VERTICAL]

VERTICAL_STIFFNESS = 170.0 \$ [N/mm] only for compatibility

VERTICAL_DAMPING = 0.0 \$ [Nms/mm] only for compatibility

\$-----OPERATING_CONDITIONS

[OPERATING_CONDITIONS]

INFLATION_PRESSURE = 2.0 \$ [bar]

TREAD_DEPTH = 8.0 \$ [mm]

AMBIENT_TEMPERATURE = 20.0 \$ [degC]

MODEL_LEVEL = 7 \$ at present only 6: standard,
\$ or 7: full 3D

\$-----MODEL

[MODEL]

PROPERTY_FILE_FORMAT = 'FTYRE'

FUNCTION_NAME = 'TYR912'

SEPARATE_ANIMATION = 1

\$-----PARAMETER

[FTYRE_DATA]

inflation_pressure = 2.0 \$ [bar]
 tyre_section_width = 195 \$ [mm]
 tyre_aspect_ratio = 65 \$ [%]
 rim_diameter = 375.6 \$ [mm]=[25.4 * 15 inch]
 rolling_circumference = 1975 \$ [mm]
 tread_width = 160 \$ [mm]
 tyre_mass = 9000 \$ [g]
 stat_wheel_load_at_10_mm_defl = 1700 \$ [N]
 stat_wheel_load_at_20_mm_defl = 3800 \$ [N]

\$

\$.....

\$the following parameters of earlier FTyre versions are obsolete

\$and should no longer be used:

\$belt_mass =

\$unloaded_radial_stiffness =

\$.....

\$

dynamic_stiffening = 0 \$ [%]
 speed_at_half_dyn_stiffening = 5.55 \$ [mm/ms]=[m/s]
 belt_extension_at_200_kmh = 1.0 \$ [%]
 interior_volume 0.03e9 \$ [mm^3]
 volume_gradient 1.0 \$ [%/mm]

\$

f1 = 62.1 \$ in-plane rotat. [Hz]
 f2 = 81.4 \$ in-plane transl. [Hz]
 f4 = 80.0 \$ out-of-plane rotat. [Hz]
 D1 = 0.05 \$ in-plane rotat. [-]
 D2 = 0.08 \$ in-plane transl. [-]
 D4 = 0.015 \$ out-of-plane rotat. [-]

\$

\$.....

\$de-comment exactly one of:

belt_in_plane_bend_stiffn = 2.0e6 \$ [N*mm^2]=[10^6*N*m^2]
 \$f5 = 70.0 \$ in-plane bending [Hz]
 \$wheel_load_at_10_mm_defl_cleat = 1400 \$ [N]

\$

```

number_belt_segments =      80  $
number_blocks_per_belt_segm = 20  $
number_tread_strips =      6  $
maximum_time_step =      0.2  $ [ms]
BDF_parameter =      0.5  $ 0.5 .. 1.0 [-]

```

A.2 Measurement data from the tyre test bench:

The characterization of the parameters FTyre requires tests of tyre on bench as for example the measure of the efforts seen in the centre of wheel during a rolling on a bar assembled on a bench to rollers (percussion).

The results of these tests will be supplied to RENAULT/NISSAN. They will be useful to fail the model of tyre, if necessary.

Annex B – Static and Dynamic charaterisation measuring condition

B.1 Renault characteristics

- Refer to test method 31-08-878, find PRAT definition in attached figure after the table below,
- Rolling measurement: **RR curve according to ISO 18164**

B.3 NISSAN Caractéristiques: Test Items and Measuring Conditions of Part for Approval

Test Items and Measuring Conditions of Parts for Approval

a) Common item

Measurement air pressure Po:	Specified in Spec Tender
Measurement load Lo:	Specified in Spec Tender
Measurement speed Vo:	Allowable speed in accordance with the speed symbol

b) Measuring conditions of items to be tested

Test item \ Test condition		Vertical load	Air pressure	Other
Static tire characteristic	1) Diagram of vertical deflection - vertical load	0 to 1.8 Lo	Po - 30 Po Po + 30	- The test shall be conducted on a flat plate. - At least 3 points shall be measured. Note 1 The lateral load of lateral deflection is 0 to 0.5 Lo. Note 2 The longitudinal load of longitudinal deflection is 0 to 0.5 Lo.
	2) Diagram of lateral deflection - lateral load	Lo		
	3) Diagram of longitudinal deflection - longitudinal load	0.5 Lo, Lo		
	4) All of 1), 2), and 3)	L ₁	P ₁	Note 1 The lateral load of lateral deflection is 0.5 L ₁ . Note 2 The longitudinal load of longitudinal deflection is 0.5 L ₁ .
Cornering characteristic	5) Diagram of cornering force -slip angle	0.5 Lo Lo 1.5 Lo	Po - 30 Po Po + 30	- Slip angle: 1° to 15° - However, the slide angle may be 15° or less when the maximum value has been measured. - At least 5 points shall be measured per 1°.
	6) Diagram of self alignment torque -slip angle			
	7) Diagram of camber thrust -slip angle			- Camber angle: 1° to 6° - At least 3 points shall be measured.
	8) All of 5), 6), and 7)	L ₁	P ₁	- Slip angle 1?? - Camber angle 1??
Other characteristic	9) Diagram of rolling resistance - running speed	Lo	Po - 30 Po Po + 30	Running speed : 30 to Vo km/h
	10) Diagram of dynamic load radius - running speed	0.5 Lo Lo 1.5 Lo		Running speed : 30 to Vo km/h
	11) All of 9), 10)	L ₁	P ₁	Running speed: 80 km/h
	12) Diagram of friction coefficient - slip ratio	Lo	Po	- Take a measurement on wet and dry road surfaces. (Enter friction coefficient of road surface in the graph.) - Take a measurement under running speed conditions of 50 and 80 km/h.
	13) RCF - conicity characteristics			- Number of test samples: 8 - For conicity of sample tires, one tire or more shall be measured for -30 or less, and one tire or more shall be measured for 30 or more. - Take a measurement by rotating in both normal and reverse directions and determine a regression line. - Take measurements at a vehicle speed of 10 km/h. - Shall measure in the ISO condition.

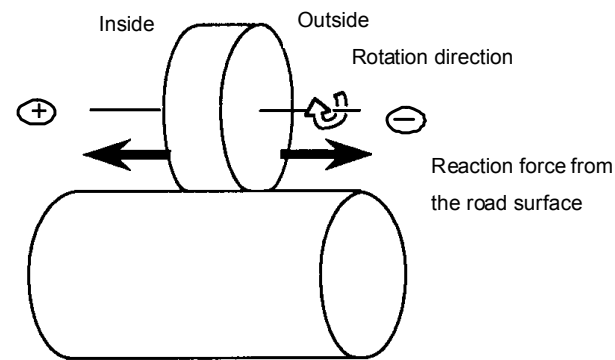
10) Diagram of dynamic rolling radius - running speed(detail)

[illegible]

c) Measuring conditions of items to be tested if needed

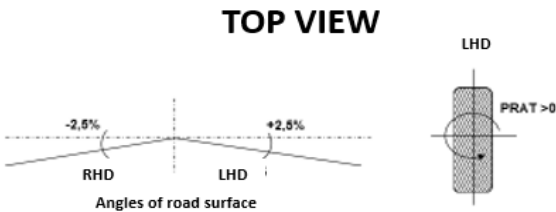
Test item	Test condition	Vertical load	Air pressure	Other
1) Diagram of transverse deflection & vertical load		0 to 1.8 Lo	Po - 30 Po Po + 30	- The test shall be conducted on a flat plate. - At least 3 points shall be measured.
2) Diagram of grounded area and vertical load				
3) Diagram of grounded width and vertical load				
4) Diagram of grounded length and vertical load				
5) Diagram of stationary steering torque and steering angle		0.5 Lo Lo 1.5 Lo	Po - 30 Po Po + 30	- Steering angle: 0° to 20° - Offset: 0
6) Threshold speed of the standing wave		Lo	Po - 30 Po Po + 30	
7) Diagram of longitudinal axial load fluctuation - speed when driving over protrusions				- Running speed: 0 to 150 km/h - Size of protrusion: 10 mm in radius, 10 mm in height, and length of grounded width or longer
8) Diagram of vertical axial load fluctuation - speed when driving over protrusions				
9) Measurement results of vehicle interior noise using a drum tester for evaluating road noise		0.5 Lo Lo 1.5 Lo	Po - 30 Po Po + 30	Running speed: 40, 60 and 80 km/h
10) Measurement results of transfer characteristics in hammering test		Lo		
11) Sound pressure level of pattern noise		Lo	Po	- Frequency: 25 to 10 kHz - Running speed: 40, 60, 80 and 100 km/h - Measuring conditions: JASO C606, bench test
12) Measurement results of force variation				- Measuring conditions: JIS D4233-01 - The average value and standard deviation of 50 samples shall be stated.

Attached Tables and Figures



Definition of direction of residual cornering force

In accordance with the standard for vehicle's right wheel, the load from the road surface, which is generated leftward against the vehicle advance direction, shall be defined as positive +.
(Same as the direction of the force of uniformity specified in JIS D4233-01/JASO C 607-00).



Definition of direction of Ply-steer Residual Auto-alignment Torque

Annex C – Electric resistance between TIRE BEADS measuring conditions

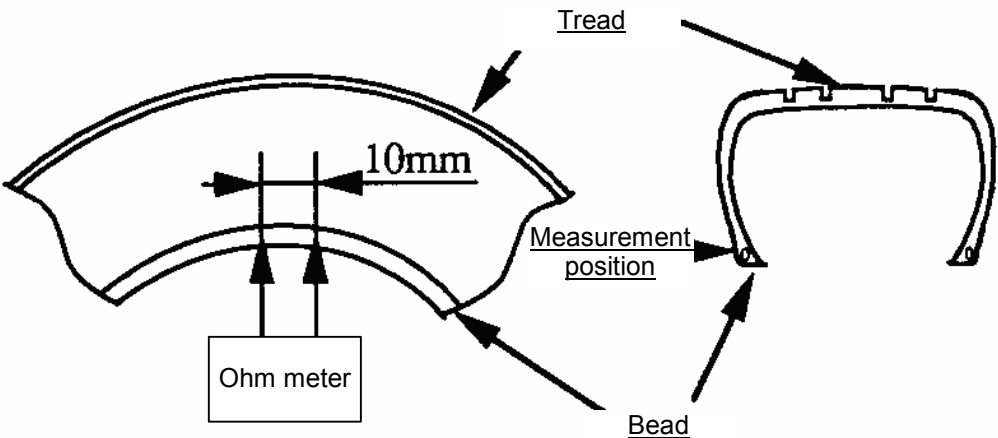


Fig. 1 (B) — Electric resistance measurement method (Tire bead (in the circumference direction))

Annex D - CHART OF THE PERTURBED FUNCTIONS OF SERVICE

TYRE

Rep.	fonct.	FONCTIONS DE SERVICE PERTURBEES <i>PERTURBED SERVICE FUNCTIONS</i> POUR LES DIFFERENTS CLIENTS CONCERNES AU COURS DE LA VIE DU COMPOSANT FOR THE VARIOUS CONCERNED CUSTOMERS DURING THE COMPONENT CYCLE LIFE	Sigle & classe Symbol & class HCPP	Objectiv e Target TNC (1) NCR (1)
		Défaut pouvant affecter la Sécurité des différents clients (Sigle S). <i>Failure which could affect the safety of various customers (S Symbol).</i> Défaut pouvant entraîner un non-respect des conditions Réglementation / Sécurité (sigle SR). <i>Failure which could lead to a non respect of Regulation / Safety requirements (SR Symbol).</i> Défaut pouvant entraîner un non-respect de la Réglementation (sigle R). <i>Failure which could lead to a regulation non respect (R Symbol).</i>	S S/SR R	0* 1.E-06 1.E-05
SR.1		Ne transmet pas les efforts entre le sol et la roue (porter, transmettre, guider, rouler et freiner). <i>Does not distribute efforts between ground and wheel (to carry, to distribute, to drive, to run, to brake).</i>		
S.1		Ne résiste pas aux pressions de gonflage maxi en utilisation . <i>Does not resist to maxi use inflating pressure.</i>		
S.2		Ne garantit pas l'étanchéité (débit de fuite > 0,1bar). <i>Does not ensure airtightness (leak rate > 0,1 bar).</i>		
S.3		Ne respecte pas l'exigence de résistivité électrique. <i>Does not meet electrical resistivity requirement.</i>		
R.1		Ne respecte pas les critères acoustiques réglementaires (bruit extérieur). <i>Does not meet acoustic regulation criteria (exterior noise).</i>		
R.2		Ne respecte pas la réglementation (en fonction des pays). <i>Does not meet regulation (according to countries).</i>		
R.3		Ne respecte pas l'environnement (profil PRV : ECE23). <i>Does not meet environment (PRV Profile : ECE23) .</i>		
R.5		Ne respecte pas la réglementation marquage (ECE30, GS-SASO 2252/2004, CCC, INMETRO). <i>Does not meet marking regulation (ECE30, GS-SASO 2252/2004, CCC, INMETRO).</i>		
R.6		N'est pas compatible avec le compteur de vitesse (circonférence de roulement). <i>Does not comply with speedometer (rolling circumference).</i>		
R.7		Ne respecte pas la réglementation recyclage. <i>Does not meet recycling regulation.</i>		
R.8		Ne respecte pas critères de résistance au roulement et de masse (émissions CO2). <i>Does not meet rolling resistance and mass criteria (CO2 emissions).</i>		
		Pour le client automobiliste, panne interdisant l'utilisation du véhicule, pour l'usine, défaut pouvant arrêter la chaîne de montage en usine (> 1 heure), ou pouvant entraîner le cafutage du véhicule par impossibilité de retouche <i>For customer, breakdown forbidding the use of the vehicle, for the factory, defect which can stop the conveyor belt in factory (> 1 hour), or being able to entail vehicle destruction by impossibility of retouch</i>	1	1.E-05
1.1		Ne résiste pas à la l'ozone. <i>Does not resist to ozone.</i>		
1.2		Ne résiste pas aux hydrocarbures. <i>Does not resist to hydrocarbon</i>		
1.3		Ne résiste pas à la température. <i>Does not resist to temperature</i>		
1.4		Ne résiste pas aux liquides de freins. <i>Does not resist to brake liquids.</i>		
1.5		Ne résiste pas aux Chlorures de Zinc et/ou Calcium <i>Does not resist to Zinc / Calcium Chloride.</i>		

STANDARD DE FONCTIONNEMENT

Continuation of the FSP table:

pour l'usine, défaut pouvant arrêter la chaîne de montage en usine (entre une 1/2 heure et une heure), ou dérive de flux (retouche lourde), ou forte dégradation des conditions de travail, pour le SAV, réparation difficile (1 journée ou plus) For the customer, defect or breakdown (not forbidding the use of the vehicle), requiring to repair quickly for the factory, the defect which can stop the conveyor belt in factory (between one half an hour and one hour), or flow drift (heavy retouch), or strong working conditions degradation for the AFTER-SALES SERVICE DEPARTMENT, difficult repair (1 day or more)		2	1.E-04
2.1	Ne satisfait pas aux exigences vibratoires (uniformité). <i>Does not comply with vibratory requirements (uniformity).</i>		
2.2	Ne satisfait pas aux exigences de tirage (Conicité, plysteer, rigidité de dérive et carossage ...) <i>Does not comply with pulling requirements (Conicity, plysteer, camber and cornering stiffness ...)</i>		
2.3	Ne respecte pas les critères de durabilité. <i>Does not meet durability criteria</i>		
2.4	Ne permet pas le montage. <i>Does not allow assembly.</i>		
2.5	Ne permet pas l'équilibrage. <i>Does not allow balancing.</i>		
2.6	Ne respecte pas le comportement routier du véhicule. <i>Does not meet vehicle road behavior</i>		
Pour le client automobiliste, défaut toléré, ou gênant mais ne nécessitant pas de réparation immédiate, pour l'usine, défaut pouvant arrêter la chaîne de montage (< 1/2 heure), ou légère retouche en ligne, ou sensible dégradation des conditions de travail pour le SAV, réparation facile (< journée) For the customer, tolerated defect, or annoying but not requiring an immediate repair, for the factory, the defect which can stop the conveyor belt (< 1/2 the hour), or light on-line retouch, or sensitive degradation of working conditions degradation for the AFTER-SALES SERVICE DEPARTMENT, easy repair (< a day)		3	2.7.E-03
3.1	Ne résiste pas aux pressions de gonflage maxi au montage. <i>Does not resist to maxi inflating pressure for assembly.</i>		
3.2	Ne respecte pas les critères acoustiques. <i>Does not meet acoustic criteria.</i>		
3.3	Ne permet pas le démontage. <i>Doesn't allow disassembly.</i>		
3.4	Ne respecte pas l'environnement (profil mini ETRTO) <i>Does not meet environment (mini profile ETRTO)</i>		
3.5	Absence de repère du sens de montage. <i>Absence of mark for mounting direction .</i>		
3.6	Ne permet pas les manœuvres aisées à l'arrêt ou à petite vitesse (parking). <i>Does not allow easy stationary or low speed (parking).</i>		
3.7	Ne respecte pas les exigences d'aspect du pneumatique <i>Does not meet tire appearance requirements</i>		

*Assurer qu'aucune non-conformité ne pourra entraîner un défaut de Sécurité ou de Réglementation dans les conditions raisonnablement prévisibles. Le fournisseur aura l'obligation de moyen suivante : justifier par des essais, que pour les caractéristiques considérées, la proportion de non-conformes pouvant entraîner le défaut de Sécurité ou de Réglementation, est inférieure à 10⁻⁶.

Annex E – Maximum Tyre Profile PPR13

PROFILE PPR 13

TECHNICAL CHARACTERISTICS OF THE OVERALL DIMENSIONS OF A TIRE WITHOUT A SUPPLEMENTARY ADHESION DEVICE

The overall dimensions of a tyre are defined according to the indications mentioned in the ETRTO standard.

1. DATA

The values to be taken into account, for a new tyre, are the ones indicated in the "DESIGN" column of the ETRTO standard.

Assume:

- DPN the diameter of the new tyre from ETRTO
- DJ the diameter of the wheel rim, (see values at the end of this appendix),
- HN the height of the section of the new tyre which is equal in $(DPN - DJ) / 2$
- LJU the width of the used wheel rim,
- LJM the width of the wheel rim of measure indicated in the E.T.R.T.O . standard,
- LPN the width of the new tyre determined on the wheel rim of measure,
- HDJ the height of the edge of the rim edge(examples: type B = 14.1 mm, type J = 17,3 mm),
- RST the radius at tyre APEX to be plotted (the values are defined at the end of this appendix),
- C1 the coefficient of calculation of the width of the tread LBR (the values are defined at the end of this appendix).

2. CONSTRUCTION OF THE MAXIMUM CONTOUR OF A TYRE

The construction of the maximum contour of a tyre is illustrated by the draft of this appendix.

The construction of the maximum contour of a tyre allows to define an outline taking into account all the tyres.

Plot the width LJU of the wheel rim and both hooks.

Calculate the width of the tyre to plot LPT according to the formula:

$LPT = [LPN + (LJU - LJM) \times 0,4] \times 1,03$ and to plot two parallel and vertical lines.

Plot the diameter DJ of the wheel rim.

Calculate the height of the section HT according to the formula:

$HT = HN \times 1.03.$

Calculate the diameter of the tyre DPT according to the formula:

$DPT = DJ + 2HT.$

Plot that diameter

Calculate the width of the tread LBR according to the formula:

$LBR = LPN \times 1,03 \times C1.$

2.1. CONSTRUCTION OF THE POINT A

The point A is defined as being the intersection of RST with the axis outlining LBR.

2.2. CONSTRUCTION OF THE AXIS XX'

The axis XX' is a horizontal axis situated in the middle of HT.

2.3. CONSTRUCTION OF THE TYRE SIDEWALL ARC OF A CIRCLE F1

Plot the arc of a circle the centre of which is situated on the axis XX'. This arc of a circle is tangent on one hand in the perpendicular at right in the axis XX' representative of the maximal width LPT and on the other hand to a circle of diameter 9 mm concentric with the circle of the rim bead (Diameter 15 mm).

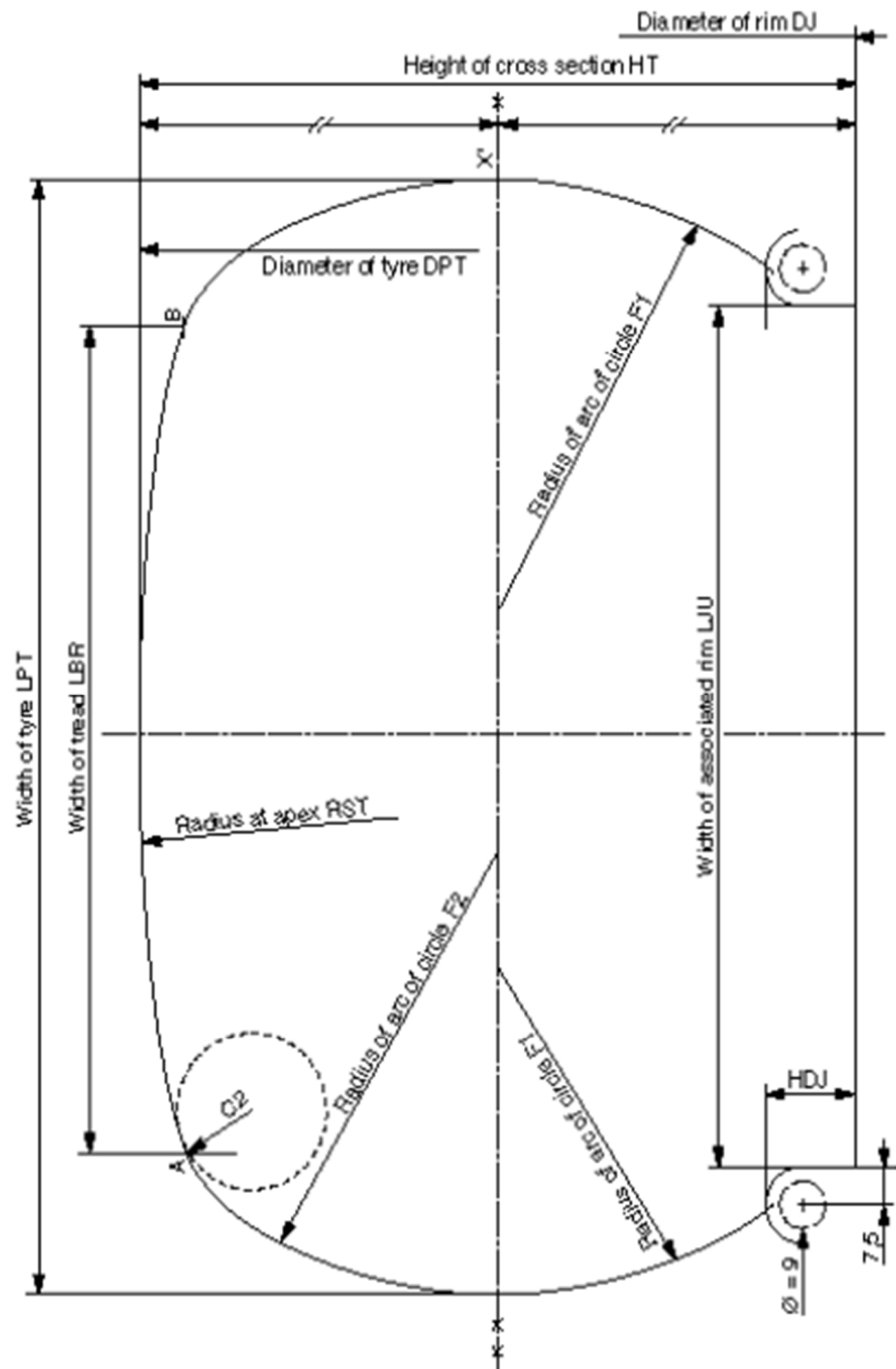
2.4. CONSTRUCTION OF THE TYRE SIDEWALL ARC OF A CIRCLE F2

Plot the arc of a circle passing by the point A whose the centre of which is situated on the axis XX'. This arc of a circle is tangent to the perpendicular to the axis XX' representative of the maximal width LPT.

2.5. CONSTRUCTION OF the "SHOULDER" PROFILE

The profile "shoulder" is an arc of a tangent circle to RST and to F2. This value varies according to the aspect ratio (see values at the end of this appendix).

EXAMPLE OF THE CONSTRUCTION OF A TYRE



VALUES OF THE TYRE APEX RADIUS RST

Largeur Rapport d'aspect	90	80	75	70	65	60	55	50	45	40	35	30	25	20
105	120	135	135	155										
115	130	150	150	170										
125	140	160	160	185										
135	155	180	180	205										
145	170	195	195	225	260	335	435	570	570	570	570	570	570	570
155	185	215	215	250	285	370	470	605	605	605	605	605	605	605
165	200	230	230	265	305	395	495	630	630	630	630	630	630	630
175	215	245	245	285	325	425	525	660	660	660	660	660	660	660
185	230	265	265	305	350	455	555	690	690	690	690	690	690	690
195	240	275	275	320	365	475	575	710	710	710	710	710	710	710
205	250	290	290	335	385	500	600	735	735	735	735	735	735	735
215	265	305	305	350	405	525	625	760	760	760	760	760	760	760
225	280	320	320	370	425	550	650	785	785	785	785	785	785	785
235	290	335	335	390	445	580	680	815	815	815	815	815	815	815
245	310	355	355	410	465	605	705	840	840	840	840	840	840	840
255	320	370	370	425	485	635	735	870	870	870	870	870	870	870
265	335	385	385	445	505	660	760	895	895	895	895	895	895	895
275	350	400	400	465	525	685	785	920	920	920	920	920	920	920
285	360	415	415	480	545	710	810	945	945	945	945	945	945	945
295	375	430	430	500	570	740	840	975	975	975	975	975	975	975
305	385	445	445	520	590	765	865	1000	1000	1000	1000	1000	1000	1000
315	400	460	460	535	610	790	890	1025	1025	1025	1025	1025	1025	1025
325	415	480	480	555	630	820	920	1055	1055	1055	1055	1055	1055	1055
335	430	495	495	575	650	845	945	1080	1080	1080	1080	1080	1080	1080
345	445	510	510	590	670	870	970	1105	1105	1105	1105	1105	1105	1105
355	455	525	525	610	690	895	995	1130	1130	1130	1130	1130	1130	1130
365	470	540	540	630	710	925	1025	1160	1160	1160	1160	1160	1160	1160
375	480	555	555	645	730	950	1050	1185	1185	1185	1185	1185	1185	1185

COEFFICIENTS C1 AND C2

PPR13 C1 TABLE

	90	80	75	70	65	60	55	50	45	40	35	30	25	20
C1	0.7	0.7	0.7	0.7	0.75	0.75	0.7	0.7	0.7	0.77	0.8	0.84	0.89	0.94

RIMS DIAMETER ACCORDING TO THE ETRTO STANDARD

TYPES OF WHEEL RIMS	DIAMETERS (mm)
10	253.2
12	304
13	329.4
14	354.8
15	380.2
16	405.6
17	436.6
18	462
19	487.4
20	512.8
21	538.2
22	563.6
23	589
24	614.4
25	639.8
26	665.2
28	716
30	766.8

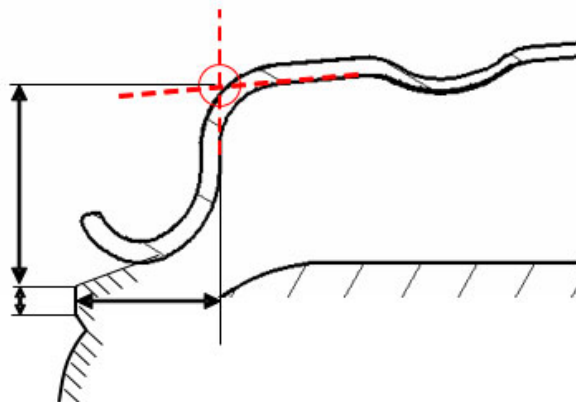
Annex F – GEOMETRY OF THE RIM PROTECTOR

B.1 Rim protector geometry

Dimensional specifications of the rim protector must be quoted on drawing with reference to ETRTO.

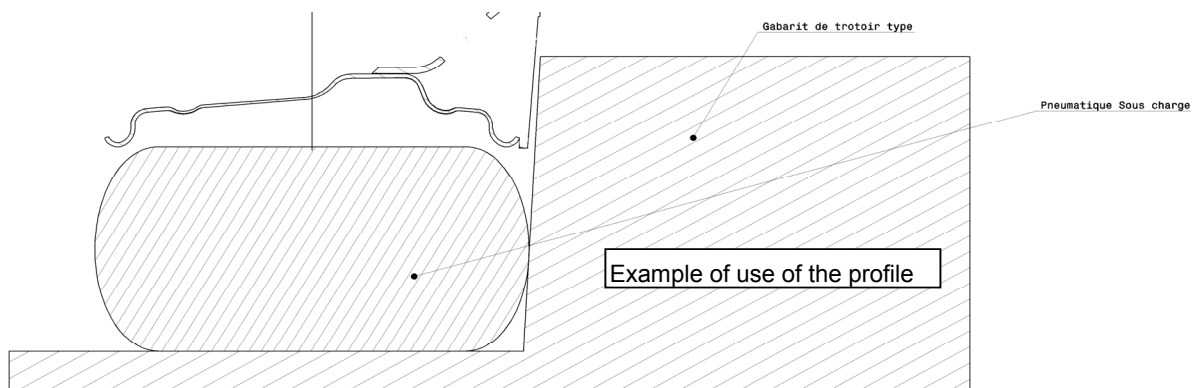
Rim protector definition must be defined according ETRTO reference point as described on picture.

Indicated dimensions can be completed by any additional characteristic necessary to precisely define geometry.



B.2 Tyre profile under load and pressure

Tyre profile under load and for given pressures should be defined on drawing and through CAD file. It should be indicated whether informations come from calculation or measurements and associated load/pressures conditions. Picture below describes the use of requested Min/Max profiles.



Annex G – REQUIREMENT FOR FLAT SPOT

Requirements	Change in high speed RFV (1st) and TFV (1st) before and after generating flat spot shall be indicated as ΔRFV and ΔTFV. When rotating continuously at high speed (100 km/h) for 30 min after generating flat spot, ΔRFV and ΔTFV shall be equal to or less than the standard tire. Note Consult with the design section in charge for the standard tire.																						
Measurement requirements (procedure)	(1) High speed uniformity measurement before generating flat spot Speed: 100 km/h Load: 80% of the allowable load at an air pressure of 200 kPa which is specified in JATMA standards Air pressure: 200 kPa (2) Generation of flat spot (2 conditions) <table border="1"> <thead> <tr> <th></th><th>A</th><th>B</th></tr> </thead> <tbody> <tr> <td>Tire warming up</td><td> Speed: 150 km/h Duration: 1 h Air pressure: Same as Section (1) Load : Same as Section (1) </td><td> Speed: 60 km/h Duration: 5 min. Air pressure: 350 kPa Load: Same as Section (1) When beginning the test immediately after the test in Section (1), warming up of the tire is not required. </td></tr> <tr> <td>Air pressure</td><td>200 kPa</td><td>350 kPa</td></tr> <tr> <td>Load</td><td>80% of 200 kPa which is specified in ETRTO standards</td><td>Maximum allowable load of tire</td></tr> <tr> <td>Generation time</td><td>16 h</td><td>720 h</td></tr> <tr> <td>Temperature</td><td colspan="2">15°C to 25°C</td></tr> <tr> <td>Tire pressing location</td><td colspan="2">Ground the tire surface opposite to the surface where low speed RFV primary peak occurs.</td></tr> </tbody> </table> (3) High speed uniformity measurement before generating flat spot (RFV, TFV) Speed, load, air pressure → same as those in Section (1) Time: 30 min. (Measurement shall be begun within 3 min after generating flat spot.) (4) Other A wheel whose primary component of the radial run out (P-P) is 0.1 mm shall be used.			A	B	Tire warming up	Speed: 150 km/h Duration: 1 h Air pressure: Same as Section (1) Load : Same as Section (1)	Speed: 60 km/h Duration: 5 min. Air pressure: 350 kPa Load: Same as Section (1) When beginning the test immediately after the test in Section (1), warming up of the tire is not required.	Air pressure	200 kPa	350 kPa	Load	80% of 200 kPa which is specified in ETRTO standards	Maximum allowable load of tire	Generation time	16 h	720 h	Temperature	15°C to 25°C		Tire pressing location	Ground the tire surface opposite to the surface where low speed RFV primary peak occurs.	
	A	B																					
Tire warming up	Speed: 150 km/h Duration: 1 h Air pressure: Same as Section (1) Load : Same as Section (1)	Speed: 60 km/h Duration: 5 min. Air pressure: 350 kPa Load: Same as Section (1) When beginning the test immediately after the test in Section (1), warming up of the tire is not required.																					
Air pressure	200 kPa	350 kPa																					
Load	80% of 200 kPa which is specified in ETRTO standards	Maximum allowable load of tire																					
Generation time	16 h	720 h																					
Temperature	15°C to 25°C																						
Tire pressing location	Ground the tire surface opposite to the surface where low speed RFV primary peak occurs.																						

Annex H – RIM SLIPPAGE

Requirements for tire rim slippage

Requirements	Tire rim slippage after testing under the following requirements shall be less than 30 mm in the circumferential direction of the rim flange.
Measurement requirements (procedure)	(1) Mount a rim with the equivalent of lubricant available at the Nissan's off-line plants applied. (The application amount of lubricant shall be the same as that of Nissan plants.)
	(2) Leave it to stand at normal temperature (15°C to 25°C) for 48 h after mounting the rim.
	(3) Note 1 After mounting to the vehicle, adjust air pressure. Then, repeat the following procedures 5 times. Setting vehicle at stationary state → Standstill acceleration at full throttle valve opening → Braking at 0.8 G from 40 km/h Note 2 Other test methods that provide same conditions may be used after consulting with design section in charge to shorten the duration.
	(4) Other Shall use the wheel which has been actually set in the Nissan vehicles.

Annex I – Red Point definition

Marking of peak point in the 1st harmonic of RFV	Shape	For plant in USA: oval shape in yellow
	Color	For plant in other areas: approximately 10 mm in dark red
	Position in radial direction	50 ± 20 mm from BEAD TOE
	Accuracy of position in circumferential direction	As per Table below

The above is basics, while the detailed final shape and color will be precised during assembly process validation.

RFV OA (Peak to Peak) [N]	Accuracy [°]
Less than 59	± 35
59 or over, less than 78	± 30
78 or over	± 25